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Evaluating the Effectiveness of Chain-of-Thought Explanations in Large Language Models: An Empirical Study of User Understanding, Trust, and Perceived Transparency

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This thesis is submitted to the Faculty of Computing at Blekinge Institute of Technology in partial fulfillment of the requirements for the degree of Masters in computer science. The thesis is equivalent to 20 weeks of full-time studies.

The authors declare that they are the sole authors of this thesis and that they have not used any sources other than those listed in the bibliography and identified as references. They further declare that they have not submitted this thesis at any other institution to obtain a degree.

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Abstract

Background. Large Language Models (LLMs) are increasingly used in applications such as education and software engineering. However, their responses can be difficult for users to understand because the reasoning behind an answer is often not visible. Chain-of-Thought (CoT) prompting provides reasoning explanations alongside LLM responses, but its effect on users' understanding, trust, and perceived transparency is not fully clear.

Objectives. This study examined how Chain-of-Thought-style explanations affected participants' perceived understanding, trust, and perceived transparency when evaluating LLM-generated responses. It also examined the relationship between perceived explanation quality and trust.

Methods. A quantitative within-participant experiment was conducted with 25 participants. Each participant evaluated standard and CoT-based responses to ten reasoning-related questions. For each response, participants rated six statements on a five-point Likert scale covering ease of understanding, understanding of how the answer was obtained, trust, confidence, transparency, and explanation clarity and usefulness. The data were analysed using descriptive statistics, the Wilcoxon signed-rank test, and Spearman's rank correlation.

Results. CoT-based responses received higher mean ratings than standard responses across all six evaluation dimensions. The overall mean rating was 4.12 for CoT responses and 2.50 for standard responses. The differences between the two conditions were statistically significant ($p < 0.001$), with large effect sizes. For RQ4, explanation quality and trust showed a positive correlation for CoT responses ($\rho = 0.37$), but the relationship was not statistically significant ($p = 0.067$).

Conclusions. The results indicate that CoT-style explanations were associated with higher self-reported perceived understanding, trust, and transparency in this study. However, the findings concern users' perceptions and do not establish that CoT explanations reveal the actual internal reasoning process of an LLM. The results also suggest that the relationship between explanation quality and trust requires further investigation with larger samples.

Data Availability Statement

No public repository accompanies this thesis. The decision not to publish the supporting material is deliberate and the reasons are set out below, together with a description of what exists and how it can be obtained.

Participant data

The survey produced one record per participant containing a submission timestamp, three background responses and 120 individual ratings. Although no directly identifying information was collected, the records are not published, for two reasons.

First, the consent text presented to participants, reproduced in Appendix A.1.1, states that responses would be used for academic research purposes. It does not state that the responses would be published, and publishing them would therefore exceed the terms under which they were given.

Second, the combination of a submission timestamp, a self-assessed experience level and an academic or professional field is potentially identifying within a small population. Participants were recruited through the author's academic and professional network, so a reader drawn from that network could plausibly associate a record with an individual. The risk is small but it is not zero, and it cannot be removed by withholding names alone.

Analysis material

The derived data on which every statistic in this thesis was computed, namely the participant-level mean for each evaluation dimension in each condition, contain no timestamps, no background information and no free text. They are retained by the author together with the analysis scripts.

Stimulus material

The ten experimental questions, the two prompts, the twenty generated responses and the complete survey instrument contain no participant data of any kind. The prompts are reproduced in full in Appendix A.2, the instrument in Appendix A.1, the questions in Table 3.2 and the final answer produced in each condition in Table A.1. The two generation sessions are preserved as shared conversation transcripts.

Availability on request

The derived analysis data, the analysis scripts and the complete set of twenty generated responses are available from the author on reasonable request for purposes

compatible with the consent under which the data were collected. Requests may be directed to the contact address on the title page.

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Contents

Abstract i

Data Availability Statement iii

Acknowledgments v

List of Abbreviations xv

1 Introduction 1
- 1.1 Introduction . . . 1
- 1.2 Background and Motivation . . . 2
- 1.3 Problem Statement . . . 3
- 1.4 Definitions of the Evaluated Concepts . . . 3
- 1.5 Research Aim . . . 4
  - 1.5.1 Objectives . . . 4
- 1.6 Research Questions . . . 5
  - 1.6.1 Hypotheses . . . 5
- 1.7 Ethical Considerations . . . 5
- 1.8 Outline of the Thesis . . . 6

2 Related Work 7
- 2.1 Related Work . . . 7
  - 2.1.1 Explainable Artificial Intelligence . . . 7
  - 2.1.2 Chain-of-Thought Prompting . . . 8
  - 2.1.3 LLM Reasoning and Evaluation . . . 8
  - 2.1.4 Synthesis and Critical Comparison of the Reviewed Work . . . 9
  - 2.1.5 Research Gap . . . 11

3 Method 13
- 3.1 Methodology . . . 13
- 3.2 Research Design . . . 13
  - 3.2.1 Experimental Design Considerations . . . 15
- 3.3 Question Formulation and Categories . . . 15
  - 3.3.1 Question Selection . . . 15
  - 3.3.2 Experimental Questions . . . 17
- 3.4 LLM Selection . . . 17
  - 3.4.1 LLM Response Generation . . . 19
  - 3.4.2 Standard Response . . . 19

3.4.3 Chain-of-Thought Response . . . 19
3.4.4 Response Review and Validation . . . 21
3.5 Survey Design . . . 21
3.6 Participants . . . 23
3.7 Experimental Procedure . . . 24
3.8 Data Collection . . . 24
3.9 Data Preparation . . . 24
3.10 Data Analysis . . . 25
3.10.1 Data Quality and Reliability . . . 26
3.11 Ethical Considerations . . . 27

4 Results and Analysis **29**
4.1 Participant Characteristics . . . 29
4.1.1 Composition of the Sample . . . 30
4.2 Reporting Conventions . . . 31
4.3 Data Structure and Aggregation . . . 31
4.4 Descriptive Results . . . 32
4.5 Interpretation of the Observed Differences . . . 33
4.5.1 The Two Conditions Fall in Different Response Categories . . . 33
4.5.2 Why Trust Showed the Smallest Difference . . . 35
4.5.3 The Ordering of the Dimensions . . . 36
4.5.4 Practical Implications for Users . . . 36
4.6 Overall Response Evaluation . . . 37
4.7 RQ1: Effect of CoT Explanations on User Understanding . . . 38
4.7.1 Answer to RQ1 . . . 38
4.8 RQ2: Effect of CoT Explanations on User Trust . . . 38
4.8.1 Answer to RQ2 . . . 38
4.9 RQ3: Effect of CoT Explanations on Perceived Transparency . . . 39
4.9.1 Answer to RQ3 . . . 39
4.10 RQ4: Relationship Between Explanation Quality and User Trust . . . 39
4.10.1 Answer to RQ4 . . . 40
4.11 Summary of Results . . . 40

5 Discussion **43**
5.1 Overview of the Main Findings . . . 43
5.1.1 Consistency of the Observed Response Pattern . . . 44
5.2 Discussion of RQ1: User Understanding . . . 44
5.3 Discussion of RQ2: User Trust . . . 45
5.4 Discussion of RQ3: Perceived Transparency . . . 46
5.5 Discussion of RQ4: Relationship Between Explanation Quality and Trust . . . 46
5.6 Relation to Previous Research . . . 48
5.6.1 Findings That Confirm Previous Research . . . 48
5.6.2 Findings That Diverge From Expectation . . . 48
5.6.3 Where the Present Study Cannot Be Compared . . . 49
5.7 Cross-RQ Discussion . . . 49
5.8 Benefits and Limitations of CoT-style Explanations . . . 50

5.9 Order Effects . . . 51
5.10 Limitations of the Trust Findings . . . 51
5.11 Implications for Human-Centred LLM Design . . . 52
5.12 Discussion Summary . . . 53

6 Conclusions and Future Work **55**
6.1 Future Work . . . 57
6.1.1 Adoption of a Validated Trust Instrument . . . 57
6.1.2 Counterbalanced Presentation and Fuller Demographics . . . 58
6.1.3 Larger and More Diverse Participant Samples . . . 58
6.1.4 Comparison of Multiple Large Language Models . . . 58
6.1.5 Evaluation of Explanation Faithfulness . . . 58
6.1.6 Different Explanation Strategies . . . 58
6.1.7 Task and Domain Variation . . . 59
6.1.8 Long-Term Trust and Reliance . . . 59
6.1.9 Uncertainty and Explanation Design . . . 59
6.1.10 Investigating the Relationship Between Explanation Quality and Trust . . . 59

References **61**

A Supplemental Information **65**
A.1 Survey Instrument . . . 65
A.1.1 Introduction and Consent . . . 65
A.1.2 Background Items . . . 65
A.1.3 Presentation of the Response Material . . . 66
A.1.4 Evaluation Items . . . 66
A.2 Response Generation Material . . . 66
A.2.1 Prompts . . . 66
A.2.2 Final Answers in Both Conditions . . . 67
A.2.3 Model and Settings . . . 67
A.2.4 Complete Response Pairs . . . 69

List of Figures

3.1 Overview of the research methodology. 14
3.2 The two response conditions compared, shown for one of the ten experimental questions. 20
3.3 Mapping between the research questions, the six evaluation dimensions, and the questionnaire items used to measure them. 23

4.1 Distribution of participant-level scores by dimension and condition ($n = 25$). 34
4.2 Mean ratings for the two conditions positioned against the verbal anchors of the response scale. 35
4.3 Mean differences between the two conditions in rank order, with the standard-condition baseline for each dimension. 36
4.4 Relationship between perceived explanation clarity and usefulness and trust for CoT-style responses ($n = 25$). 40

5.1 Non-causal summary of the main observed differences and association identified in the study. Dashed lines indicate observed relationships and should not be interpreted as causal pathways. 50

List of Tables

2.1 Critical comparison of the reviewed literature 10

3.1 Reasoning categories used for question formulation 16
3.2 Questions used in the experiment 18
3.3 Validation of standard and CoT responses 21
3.4 Cronbach's alpha for the paired items 26

4.1 Participant characteristics 30
4.2 Comparison of CoT-style and standard responses 32
4.3 Wilcoxon signed-rank tests comparing the two conditions 33
4.4 Exploratory overall evaluation of CoT-style and standard responses . 37
4.5 User understanding under CoT-style and standard conditions 38
4.6 Spearman correlation between explanation clarity and usefulness and trust 39
4.7 Summary of findings in relation to the research questions 41

A.1 Final answer produced in each condition for the ten experimental questions 68
A.2 Generation configuration 68

List of Abbreviations

Abbreviation	Expansion
AI	Artificial Intelligence
API	Application Programming Interface
BTH	Blekinge Institute of Technology
CoT	Chain-of-Thought
GPT	Generative Pre-trained Transformer
LLM	Large Language Model
RQ	Research Question (RQ1–RQ4)
SD	Standard Deviation
XAI	Explainable Artificial Intelligence

Notation used in the statistical analysis

Symbol	Meaning
n	Number of participants ($n = 25$ throughout)
$H1$–$H4$	Hypotheses corresponding to RQ1–RQ4
Z	Standardised test statistic of the Wilcoxon signed-rank test
p	Probability value associated with a test statistic
r	Effect size for the Wilcoxon signed-rank test, $r = \lvert Z \rvert / \sqrt{N}$
ρ	Spearman's rank correlation coefficient
α	Significance level, set at 0.05

Note. All six evaluation dimensions were measured on a five-point Likert scale, where 1 denotes *strongly disagree*, 3 denotes *neither agree nor disagree*, and 5 denotes *strongly agree*. Ratings were averaged across the ten questions for each participant and condition before analysis, so every reported statistic is based on 25 participant-level scores rather than on individual ratings.

Chapter 1

Introduction

1.1 Introduction

Large Language Models (LLMs) have become an important part of Artificial Intelligence in recent years. Models such as GPT, Gemini, Claude, and LLaMA are based on transformer architectures and have improved the ability of AI systems to understand and generate natural language [1, 9, 10]. LLMs are now used in many different areas, including chatbots, education, writing assistance, programming, and decision-support applications.

Although LLMs can produce useful and fluent answers, it is often difficult for users to understand how an answer was produced. The model usually presents the final response without showing the steps or information that led to it. This has raised concerns about transparency, accountability, and interpretability in AI systems [11, 12]. For users, this can also make it harder to judge whether an answer is reliable and whether the system should be trusted.

Explainable Artificial Intelligence (XAI) focuses on making AI systems easier for people to understand and interpret [4, 13]. In the case of LLMs, Chain-of-Thought (CoT) prompting is one approach that can be used to provide more information about how an answer is reached.

Chain-of-Thought prompting asks the model to generate intermediate reasoning steps before providing the final answer [2, 3]. As a result, the user receives more than just the final answer and can also see an explanation of the reasoning presented by the model.

Previous research has mainly examined whether Chain-of-Thought prompting improves the performance of LLMs on reasoning tasks. Less attention has been given to the user's perspective. In particular, it is not clear whether users find LLM responses easier to understand when an explanation is included, whether the explanation affects their trust in the answer, or whether it makes the system appear more transparent.

The growing use of LLMs in areas such as education, healthcare, and software engineering makes these questions increasingly relevant. When people use AI systems to support decisions or complete tasks, they need to be able to understand and evaluate

the information provided by the system. Previous research on explainable AI has therefore emphasised the importance of explanations, transparency, and user understanding [4, 13, 14].

This thesis examines Chain-of-Thought explanations from a user centred perspective. It compares standard LLM responses with responses that include Chain-of-Thought explanations and focuses on three main aspects of the user experience: understanding, trust, and perceived transparency.

1.2 Background and Motivation

Artificial Intelligence is becoming part of many activities that people perform in their everyday and professional lives. Large Language Models are now used in areas such as education, software engineering, customer support, and research [9, 10].

At the same time, LLMs do not always provide correct information. They can produce hallucinated or misleading answers while still presenting the information in a fluent and convincing way [11]. This makes it difficult for users to judge the reliability of an answer, especially when they do not have enough knowledge to verify it themselves.

Chain-of-Thought prompting is interesting in this context because it provides additional information between the question and the final answer. Previous studies have shown that Chain-of-Thought prompting can improve performance on a range of reasoning tasks [2,3]. However, better reasoning performance does not necessarily mean that users understand the response better or trust it more.

For this reason, this study looks at the user side of LLM explanations rather than only the performance of the model. The motivation is to understand whether providing a reasoning explanation changes how people evaluate an AI-generated answer.

Trust is an important part of human-AI interaction. Users need to have an appropriate level of trust when relying on AI-generated information. Previous research has linked trust in AI systems with factors such as transparency, explainability, perceived competence, and reliability [5, 14]. In this study, transparency refers to the extent to which users feel that they can understand how the system arrived at its answer [13, 14].

Wei et al. [2] showed that Chain-of-Thought prompting can encourage language models to produce intermediate reasoning steps before giving an answer. Similarly, Kojima et al. [3] showed that a simple instruction such as "Let's think step by step" can improve the reasoning performance of language models. These studies provide the basis for using CoT in the present study. However, rather than measuring whether CoT improves the model's task performance, this thesis focuses on how users respond to the explanations that are presented to them.

1.3 Problem Statement

Large Language Models are increasingly used for tasks that involve reasoning, explanation, and decision support. However, a user often sees only the final answer and has limited information about how the model arrived at it. This can make it difficult to judge the response and may affect how much the user trusts the system.

Chain-of-Thought prompting provides a way of presenting intermediate reasoning information together with the final answer. Although previous research has examined the effect of CoT on model performance, there is less empirical evidence about how users perceive these explanations.

The main problem addressed in this thesis is the limited empirical understanding of whether Chain-of-Thought explanations improve user understanding, trust, and perceived transparency when interacting with LLM-generated responses.

Three specific things are absent from the existing literature, and each of them corresponds directly to a research question in this thesis. First, work on CoT prompting evaluates reasoning steps through task accuracy [2, 3] and does not ask whether a reader understands the answer better when those steps are shown; RQ1 addresses this. Second, the XAI literature establishes that explanation supports appropriate reliance [16, 25], but studies this for feature-based explanations of classical models rather than for natural-language reasoning produced by an LLM; RQ2 and RQ3 address trust and perceived transparency in that setting. Third, although explanation quality and trust are often discussed together, the relationship between them is rarely measured directly, and Turpin et al. [26] give theoretical reason to expect them to come apart, since an explanation may be judged clear without being faithful to the process that produced the answer; RQ4 addresses this relationship empirically.

The contribution of this thesis is therefore to move the evaluation of CoT from the model to the reader: to measure what changes in a user's understanding, trust and perceived transparency when the same model, answering the same questions, presents its reasoning rather than withholding it.

1.4 Definitions of the Evaluated Concepts

Because the four concepts examined in this thesis are closely related and are used inconsistently in the literature, they are defined here in the specific sense in which they are measured in this study. All four are perception measures obtained by self-report.

Understanding

The extent to which a participant reports being able to follow a response and to see how its answer was obtained. Two items were used, separating ease of understanding the response from understanding the process behind the answer. This is perceived understanding; no comprehension test was administered.

Trust The extent to which a participant reports being willing to accept and rely on an answer. Measured separately from confidence, which concerns willingness to act on the answer in a specific instance. Neither item measures whether the answer merited that reliance.

Perceived transparency
The extent to which a participant feels able to see how the system arrived at its answer. This is distinct from faithful or technical transparency, which would require the explanation to represent the computation that produced the answer. Following Turpin et al. [26], the two are treated as separable throughout this thesis.

Explanation quality
The extent to which a participant judges an explanation clear and useful. Captured by a single item, and therefore referred to as perceived explanation clarity and usefulness rather than as a validated measure of explanation quality.

The four are examined separately because there is no reason to assume they move together. An explanation may be clear without being trusted, and a response may appear transparent without being faithful. Whether they do move together is in part an empirical question, and RQ4 addresses one instance of it.

1.5 Research Aim

The aim of this thesis is to evaluate the effectiveness of Chain-of-Thought explanations in Large Language Models by investigating their impact on user understanding, trust, and perceived transparency.

1.5.1 Objectives

The objectives of this thesis were:

1. To compare participants' perceived understanding of LLM responses with and without CoT-style explanations.

2. To compare participants' self-reported trust and confidence between the two response conditions.

3. To examine participants' perceived transparency of responses presented with and without CoT-style explanations.

4. To examine the relationship between perceived explanation quality and user trust.

1.6 Research Questions

The thesis address the following research questions:

1. How do Chain-of-Thought explanations influence user understanding of LLM-generated responses?
2. Do Chain-of-Thought explanations improve user trust in LLM-generated answers?
3. How do users perceive the transparency of LLM responses when explanations are provided?
4. What relationship exists between explanation quality and user trust?

1.6.1 Hypotheses

To guide the empirical analysis, the following hypotheses were considered:

- **H1:** Participants report higher perceived understanding for LLM responses with CoT-style explanations than for standard responses.
- **H2:** Participants report higher trust in LLM responses with CoT-style explanations than in standard responses.
- **H3:** Participants report higher perceived transparency for LLM responses with CoT-style explanations than for standard responses.
- **H4:** Perceived explanation quality be positively associated with user trust.

1.7 Ethical Considerations

Participation in the study was voluntary. Before completing the questionnaire, participants were given information about the purpose of the study, the evaluation procedure, and how their responses would be used. They were also informed that participation was optional and that they could stop participating without any negative consequences.

No personally identifiable information was collected through the questionnaire. The responses were treated as confidential and were analysed in an aggregated form. The collected data were used only for this research, and individual participants were not identified in the analysis.

The study did not involve sensitive personal questions or activities expected to cause physical or psychological harm. Participants were only asked to evaluate AI-generated responses and provide their opinions about understanding, trust, transparency, and the quality of the explanations.

The participants were not explicitly told which responses were standard responses

and which contained Chain-of-Thought explanations. This was done to reduce the possibility that knowing the response type would influence their ratings. The two response conditions were presented as part of the same evaluation task so that participants could focus on the responses themselves rather than on the label of the condition.

All collected responses were stored securely and were accessible only to the researcher. The results were reported in aggregated form, so individual participants could not be identified from the thesis.

1.8 Outline of the Thesis

The remainder of this thesis is organised as follows. Chapter 2 presents the related work on explainable AI, Chain-of-Thought prompting, and user-centred evaluation of AI explanations. Chapter 3 describes the research methodology, including the experimental design, response generation, questionnaire, participants, data collection, and analysis methods. Chapter 4 presents the results of the user evaluation and answers the four research questions. Chapter 5 discusses the findings in relation to previous research and the limitations of the study. Chapter 6 concludes the thesis and presents suggestions for future work.

Chapter 2

Related Work

2.1 Related Work

2.1.1 Explainable Artificial Intelligence

Angelov et al. [25] provided an analytical review of Explainable Artificial Intelligence (XAI), covering the development of explainability methods, their main principles, and the challenges involved in making complex AI systems easier to understand. The authors also discussed why explainability and transparency are important when AI systems are used in practice. This provides a useful foundation for this thesis, which looks at whether explanations generated by Large Language Models (LLMs) can help users understand responses and perceive them as more transparent. However, the work by Angelov et al. [25] considers AI systems more generally, while this thesis focuses on natural-language explanations produced using Chain-of-Thought (CoT) prompting.

Das [16] and Rad [16] reviewed different approaches to Explainable Artificial Intelligence and discussed the problems caused by the black-box nature of many machine learning models. Their survey covers several XAI techniques and points out the role of interpretability in building AI systems that people can trust. This is related to the present study because understanding an AI system and trusting its output are closely connected to the way explanations are presented. However, their work mainly discusses traditional machine learning and deep learning models rather than explanations produced by LLMs. This thesis applies the same general idea of explainability to LLM-generated responses and examines how users evaluate them.

Dwivedi et al. [20] reviewed a range of XAI techniques, frameworks, and tools used to develop explainable AI systems. The authors discussed the role of explanations in helping users understand AI decisions and in supporting appropriate trust. This is relevant to the current study because it highlights that explainability should also be considered from the user's point of view. At the same time, the review covers many different AI models and explanation methods, whereas this thesis is concerned with one specific approach, namely CoT explanations generated by LLMs.

2.1.2 Chain-of-Thought Prompting

Wang et al. [17] examined which factors contribute to the effectiveness of Chain-of-Thought prompting in Large Language Models. Using experiments and ablation studies, they found that CoT could still work even when some demonstrations included invalid reasoning steps. Their results suggested that factors such as the relevance and order of reasoning steps can affect the performance of CoT. This is important for the present study because it shows that the way reasoning information is presented can matter. However, Wang et al. mainly evaluated CoT based on the reasoning performance of the model. The current thesis takes a different approach by looking at how users respond to the explanations.

Zhang et al. [19] proposed Automatic Chain-of-Thought (Auto-CoT), which automatically creates CoT demonstrations instead of requiring them to be manually written. Their experiments on several reasoning benchmarks showed that Auto-CoT could achieve performance comparable to, and in some cases better than, manually designed CoT prompts. The study therefore provides evidence that automatically generated reasoning chains can be useful for LLM reasoning. However, the evaluation was based mainly on benchmark performance. It did not examine whether people found these reasoning chains understandable or trustworthy. The present thesis focuses on this user perspective.

Stechly et al. [22] studied the limitations of CoT prompting in planning tasks. Their results showed that the effectiveness of CoT can depend on factors such as prompt specificity and task complexity. They also found that improvements obtained with CoT do not necessarily generalise to more complex tasks. This is relevant to the current study because it shows that simply adding reasoning steps does not guarantee that the reasoning itself is reliable. An explanation can look detailed and convincing while still containing errors. For this reason, the present study considers not only whether an explanation is provided, but also how users evaluate it and whether it affects their trust.

2.1.3 LLM Reasoning and Evaluation

Yehudai et al. [18] reviewed different approaches for evaluating LLM-based agents. Their survey covers areas such as reasoning, planning, tool use, application-specific benchmarks, and evaluation frameworks. The authors argue for more detailed evaluation methods rather than relying on a single measure of system performance. This broader view of evaluation is relevant to this thesis because the current study also goes beyond model performance and considers user responses. However, their main focus is on LLM-based agents, while this thesis evaluates an explanation method using human participants.

Xu et al. [24] reviewed tool-learning approaches for LLMs, including tool retrieval, task decomposition, planning, tool selection, and tool invocation. They also discussed evaluation measures and safety issues related to LLM-based systems. This study is not directly concerned with explainability or user trust, so its relevance to

the current thesis is limited. It is included mainly to provide broader context on how LLM systems are evaluated and used.

Wu et al. [21] reviewed the development and applications of LLM fine-tuning and discussed several challenges associated with the use of these models. The review also considers reliability, credibility, adaptability, and human–AI collaboration. Although the study does not specifically examine CoT explanations, the discussion of reliability and human–AI interaction is relevant to the broader problem addressed in this thesis. The current study narrows this issue down to whether explanations affect how users understand and trust LLM responses.

Li et al. [23] proposed SnapKV, a method for improving the memory and computational efficiency of LLMs when working with long-context inputs. Their experiments showed improvements in generation efficiency while maintaining comparable model performance. The main focus of this work is computational efficiency rather than explainability or user interaction. Therefore, it has only a limited connection to the research questions in this thesis, but it illustrates the wider development of techniques for improving LLM systems.

2.1.4 Synthesis and Critical Comparison of the Reviewed Work

The studies reviewed above differ in what they measure, in the evidence they provide, and in what they leave unresolved. Table 2.1 summarises these differences, and the comparison below draws out three points of tension that motivate the present study.

The first tension is between performance and comprehension. Wei et al. [2] and Kojima et al. [3] both demonstrate accuracy gains, and both treat the reasoning trace as an internal mechanism rather than as something a person reads. Wang et al. [17] sharpen the problem considerably by showing that the accuracy benefit survives when the reasoning steps are made invalid, provided they remain relevant and correctly ordered. If a reasoning trace can be logically wrong and still improve model output, then its effect on a human reader cannot be inferred from the accuracy figure at all.

The second tension concerns what an explanation is evidence of. The XAI literature [16, 20, 25] generally treats an explanation as a window onto a model's decision process, and evaluates it accordingly. Turpin et al. [26] show that this assumption does not hold for CoT: a model can state reasoning that did not produce its answer. The two literatures therefore disagree about what user-facing explanation actually demonstrates, and this disagreement is what makes the distinction between perceived and faithful transparency essential in the present study rather than merely cautious.

The third tension is methodological. Stechly et al. [22] find that CoT fails to generalise to planning, while the studies above report consistent gains on arithmetic and commonsense reasoning. Yehudai et al. [18] argue that this inconsistency is partly an artefact of evaluating complex systems through single performance measures. Taken together, these positions suggest that CoT should be evaluated on several dimensions at once and in relation to a specified task type, which is the approach adopted in this thesis: six evaluation dimensions applied to ten reasoning questions of stated categories.

Table 2.1: Critical comparison of the reviewed literature

Work	**What is measured**	**What it establishes**	**What it leaves open**
Wei et al. [2]	Task accuracy on reasoning benchmarks	CoT prompting elicits intermediate reasoning and raises accuracy	Whether a reader benefits from seeing those steps
Kojima et al. [3]	Zero-shot accuracy	A single instruction is sufficient to elicit reasoning	Same limitation; no user measurement
Wang et al. [17]	Accuracy under perturbed prompts	Validity of reasoning matters less than its relevance and ordering	Implies reasoning may persuade without being correct; untested with users
Zhang et al. [19]	Accuracy with automatically constructed prompts	CoT can be generated without manual exemplars	Explanation quality as experienced by a reader
Stechly et al. [22]	Planning task performance	CoT does not generalise to planning	Where the boundary of usefulness lies for users
Turpin et al. [26]	Faithfulness of stated reasoning	Explanations can rationalise an answer post hoc	Whether readers can detect this; whether unfaithful explanations still raise trust
XAI reviews [16,20,25]	Interpretability methods and user studies	Explanation supports understanding and calibrated reliance	Mostly feature-based explanations of classical models, not LLM natural language
Yehudai et al. [18]	Evaluation practice for LLM agents	Single performance measures are inadequate	Which user-facing measures should replace them

2.1.5 Research Gap

The literature reviewed above covers several areas related to this thesis, including Explainable Artificial Intelligence, natural-language explanations, Chain-of-Thought prompting, and the evaluation of LLM systems. XAI research has established the importance of explainability and transparency when people interact with AI. Research on CoT has, in comparison, mainly examined whether reasoning steps help LLMs perform better on reasoning tasks.

A large part of the CoT literature uses technical measures such as accuracy, benchmark performance, and reasoning performance. These measures are useful for evaluating the model, but they do not tell us how users experience the explanations. Similarly, research in XAI has examined trust and user understanding, but much of this work has focused on traditional machine learning and deep learning systems rather than natural-language explanations produced by LLMs.

This leaves a gap in understanding how users perceive CoT explanations in LLM responses. In particular, there is a need to examine whether adding a reasoning explanation changes users' understanding of the answer, their trust in the response, and their perception of the transparency of the system.

This thesis addresses this gap through a user-centred experimental study. Standard LLM responses and responses containing CoT explanations are presented to participants, who evaluate both conditions using a questionnaire. The collected responses are then analysed quantitatively to compare the two conditions and to examine the relationship between explanation quality and trust. In this way, the study adds a human-centred perspective to existing research that has largely evaluated CoT through model performance.

Chapter 3

Method

3.1 Methodology

This chapter describes the methodology used to investigate how Chain-of-Thought (CoT) explanations affect users' understanding, trust, and perceived transparency when interacting with Large Language Models (LLMs). The study used a quantitative, within-participant design. The same LLM was used to generate two types of responses for the same set of questions: standard responses and responses containing CoT explanations.

Participants evaluated both types of responses using a structured questionnaire. The methodology covered the selection of questions, generation and validation of the LLM responses, survey design, participant recruitment, data collection, data preparation, and statistical analysis.

3.2 Research Design

The study used a quantitative within-participant experimental design. The main purpose was to compare participants' evaluations of standard LLM responses with their evaluations of responses that included CoT explanations.

For each question, two responses were generated using the same LLM. One response provided a direct answer without a detailed reasoning explanation, while the other provided an answer together with a step-by-step explanation.

Each participant evaluated both response conditions. This made it possible to compare the two conditions within the same participant instead of comparing two separate groups of participants. Using this design also reduced the effect of individual differences, such as previous experience with AI tools.

The two conditions were presented in a fixed order. The questionnaire contained two labelled blocks, Response A followed by Response B, and every participant encountered them in that sequence. The labels carried no indication of which block contained which condition, so participants remained blind to the manipulation, but the order itself was not varied. The consequences for interpretation are discussed in Section 5.9. Participants were not explicitly told which response represented the standard condition and which represented the CoT condition.

The overall research procedure is shown in Figure 3.1.

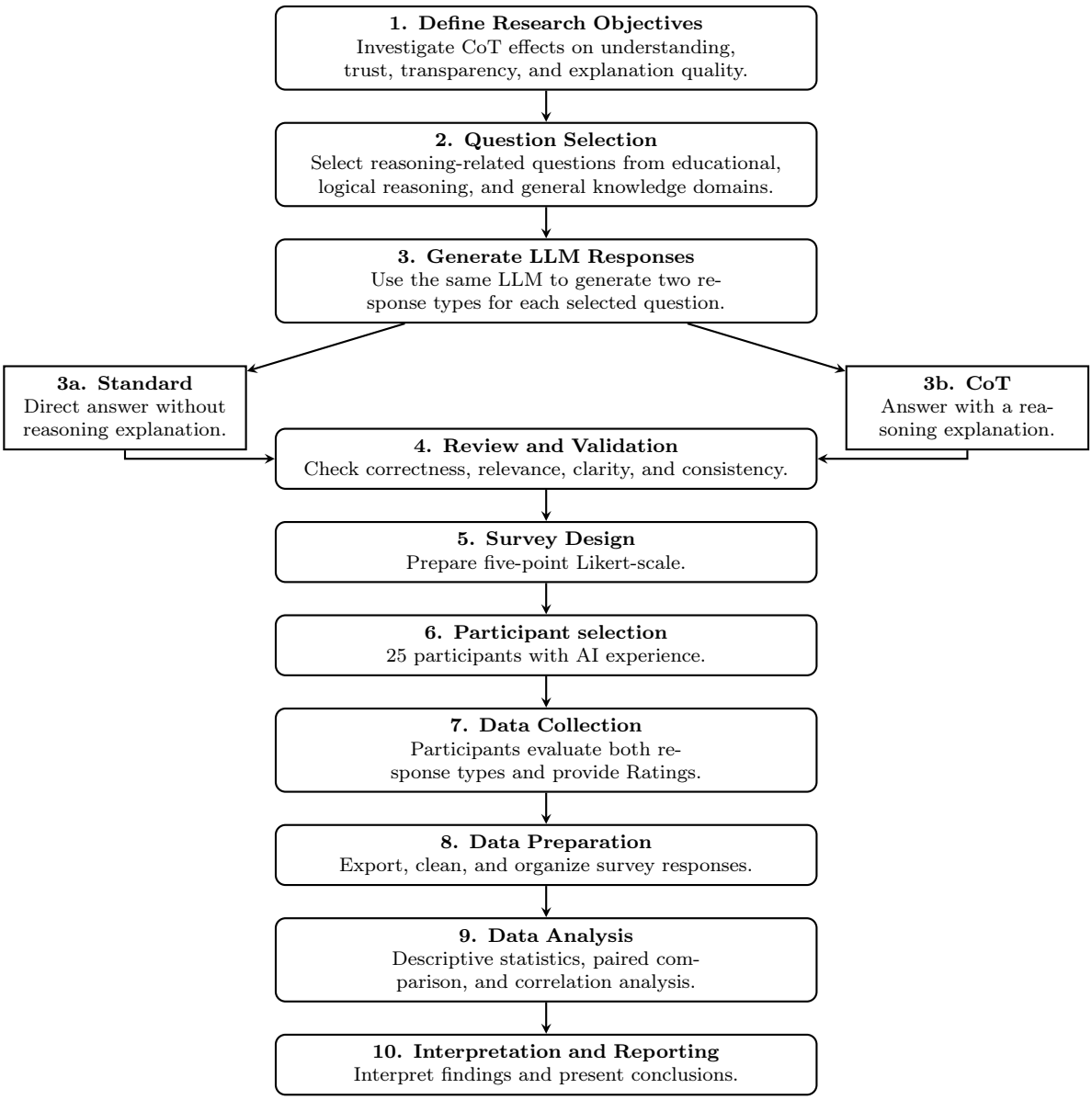


Figure 3.1: Overview of the research methodology.

3.2.1 Experimental Design Considerations

The two response conditions used the same questions and the same Large Language Model, but they differed in more than the explicit presence of an explanation. The standard responses were instructed to be concise, whereas the CoT-style responses were instructed to provide a step-by-step explanation. As a result, the CoT-style responses were generally longer and contained more information and detail.

Therefore, the experimental design does not completely isolate the contribution of Chain-of-Thought reasoning from other characteristics of the response format, such as response length, information content, level of detail, and perceived effort. The comparison should consequently be interpreted as a comparison between standard concise responses and CoT-style step-by-step responses rather than as an isolated causal test of Chain-of-Thought reasoning.

3.3 Question Formulation and Categories

The questions were selected to include different types of reasoning rather than focusing on a single type of task. The categories were based on common reasoning tasks discussed in the literature, including arithmetic reasoning, logical reasoning, multi-step reasoning, commonsense reasoning, and symbolic reasoning.

Since the study focused on the user experience of LLM explanations, the questions also needed to be suitable for evaluating whether participants could understand and assess the responses.

The questions were designed to require some reasoning, interpretation, or decision-making rather than simple factual recall. They were also kept understandable for participants from different academic and professional backgrounds. The same questions were used in both experimental conditions so that the question itself did not change between the standard and CoT responses.

The main categories used for question formulation are presented in Table 3.1.

An initial question bank be developed across these categories. The questions are then be reviewed based on clarity, relevance, reasoning complexity, and suitability for both experimental conditions. Questions that are overly simple, require specialised knowledge, or depend on rapidly changing information is be excluded. The final questions provide sufficient variation in reasoning type while keeping the survey manageable for participants.

3.3.1 Question Selection

The questions were selected from several reasoning-oriented areas, including mathematical reasoning, logical reasoning, commonsense reasoning, deductive reasoning, and scenario-based decision-making.

The following criteria were used when selecting the questions:

Table 3.1: Reasoning categories used for question formulation

Category	**Description**
Mathematical Reasoning	Problems involving calculations, percentages, probability, or numerical relationships.
Logical Reasoning	Problems requiring evaluation of whether a conclusion follows from given statements or facts.
Deductive Reasoning	Problems requiring specific conclusions to be derived from a set of known facts or premises.
Social and Interpersonal Reasoning	Situations involving the behaviour of other people, where several explanations are possible and the appropriate response depends on judgement rather than calculation.
Commonsense Reasoning	Problems requiring the application of everyday knowledge to interpret a situation or determine a plausible outcome.
Multi-Step Reasoning	Problems requiring several reasoning steps or the combination of multiple pieces of information to reach an answer.
Scenario-Based Reasoning	Realistic situations requiring interpretation of information and selection of an appropriate response.
Decision-Making	Situations involving multiple alternatives where the response requires comparison of options and justification of a decision.
Cause-and-Effect Reasoning	Problems requiring evaluation of whether one event or observation can reasonably be considered a cause of another.
Ambiguous and Uncertain Situations	Questions where the available information may be insufficient for a definite conclusion, requiring the model to recognise and communicate uncertainty.

- The question should have a clear or identifiable answer.
- The question should require some level of reasoning.
- The question should be understandable without specialised knowledge.
- The question should allow both a direct answer and an explanation.
- The question should not depend on rapidly changing information.

Questions requiring highly specialised knowledge or current external information were avoided. This was done to reduce the influence of differences in participants' individual subject knowledge.

Ten questions were selected for the final experiment. The number was kept relatively small so that participants could evaluate both response conditions without making the questionnaire unnecessarily long.

3.3.2 Experimental Questions

Ten questions were used in the final experiment. They covered different types of reasoning, including mathematical, logical, commonsense, deductive, social and interpersonal, cause-and-effect, scenario-based, ambiguous, and risk-evaluation reasoning.

The same set of questions was used for both experimental conditions. Each question was given to the same Large Language Model using two different prompting conditions: a standard prompt and a Chain-of-Thought (CoT) prompt. Keeping the questions unchanged allowed the response conditions to be compared more directly.

Table 3.2 presents the ten questions used in the experiment.

3.4 LLM Selection

A single Large Language Model was used to generate both response conditions. GPT-5 mini was used to generate the responses for both conditions, through the ChatGPT web interface rather than the OpenAI API. The same model was used throughout, and the two conditions differed only in the instruction given, as set out in Sections 3.4.2 and 3.4.3.

Because generation was carried out through the web interface, the decoding parameters that an API call would expose, such as temperature, top-p and maximum output length, were neither set nor recorded, and they cannot be recovered retrospectively. The model therefore operated under whatever defaults the interface applied at the time of generation. This limits reproducibility in one specific respect: a later attempt to regenerate the responses with the same prompts would not be guaranteed to produce identical text, even with the same model. What is reproducible, and what the analysis depends on, is the stimulus material itself: the twenty responses actually shown to participants are fixed, and they are retained in the form in which

Table 3.2: Questions used in the experiment

ID	Category	Question
Q1	Mathematical Reasoning	A bat and a ball cost €1.10 in total. The bat costs €1.00 more than the ball. How much does the ball cost?
Q2	Logical Reasoning	All researchers in a laboratory have access to the laboratory's database. Some people who have access to the database are interns. Can we conclude that some researchers in the laboratory are interns? Explain your answer.
Q3	Commonsense / Situational Reasoning	A doctor gives a patient three tablets and tells the patient to take one tablet every 30 minutes. How long will it take the patient to take all three tablets? Explain your answer.
Q4	Social and Interpersonal Reasoning	A colleague who is normally very talkative becomes unusually quiet during a team meeting. What are some possible reasons for this behaviour, and how should you respond appropriately?
Q5	Multi-Step Mathematical Reasoning	A jacket is originally priced at €200. The store offers a 25% discount. During a special promotion, an additional 10% discount is applied to the discounted price. What is the final price of the jacket?
Q6	Deductive Reasoning	Every employee who has completed the security training is allowed to access the company's internal system. Daniel is not allowed to access the internal system. Can we conclude that Daniel has not completed the security training? Explain your answer.
Q7	Cause-and-Effect Reasoning	A company introduces a new software system, and employee productivity decreases during the following month. Can the company conclude that the new software system caused the decrease in productivity? What other factors should be considered?
Q8	Scenario-Based Decision-Making	You are managing a software project that is one week away from its deadline. Two team members become unavailable, and three important tasks are still incomplete. What should the project manager consider when deciding which tasks to prioritize?
Q9	Ambiguous / Uncertain Reasoning	John went to the bank after work and returned home two hours later than usual. What was the reason for John's delay?
Q10	Risk and Evidence Evaluation	A social media post claims that a new medicine can completely prevent a particular disease. What information should you consider before deciding whether the claim is trustworthy?

they were generated. The two generation sessions are preserved as shared conversation transcripts, and the responses are available on request as described in the data availability statement.

The purpose of the study was not to compare different LLMs. Instead, the study focused on comparing how users evaluated standard responses and CoT-style responses generated by the same model. Using a single model provided a more controlled basis for comparing the two response conditions.

3.4.1 LLM Response Generation

For each of the ten questions, two responses were generated using the same LLM. The main difference between the two conditions was the instruction given to the model about how the answer should be presented.

3.4.2 Standard Response

For the standard condition, the LLM was instructed to provide a direct and concise answer without a detailed reasoning explanation.

The prompt used was:

> You are participating in an academic research experiment on AI-generated responses.
>
> Answer the following question clearly and concisely.
>
> Requirements:
>
> 1. Provide the final answer directly.
> 2. Do not provide a step-by-step reasoning process.
> 3. Do not provide an extended explanation.
> 4. Do not introduce information that is not required to answer the question.
>
> Question: [question text]

This condition represents a typical interaction in which the user receives the answer without an explicit reasoning explanation. The instruction is worth noting in full because the brevity of the standard responses was imposed by three explicit prohibitions rather than arising incidentally: the model was told not to reason step by step, not to explain at length, and not to add unrequired information. The consequences of this for the comparison are discussed in Section 3.2.1. The complete prompts for both conditions are reproduced in Appendix A.2.

3.4.3 Chain-of-Thought Response

For the CoT condition, the same LLM was given the same question but was instructed to provide a clear step-by-step explanation before presenting the final answer.

The prompt used was:

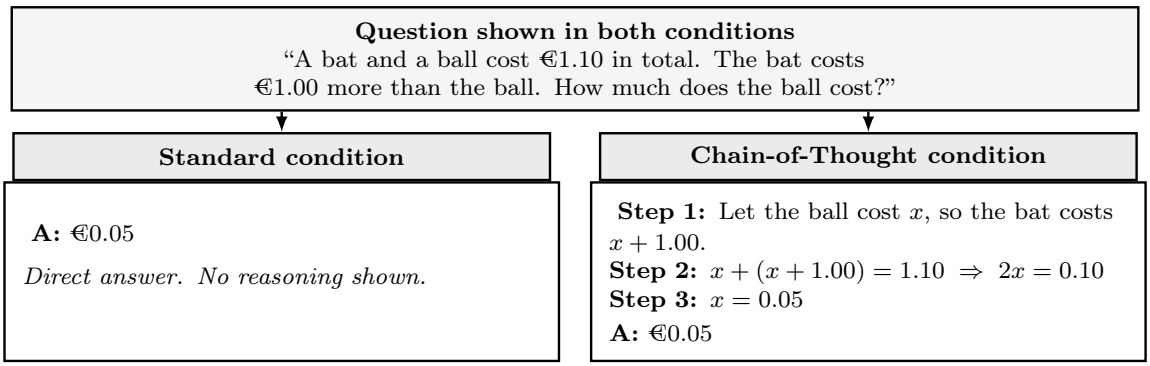
> You are participating in an academic research experiment on AI-generated responses.
>
> Answer the following question and provide a clear, step-by-step explanation that helps a user understand how the answer was obtained.
>
> Requirements:
>
> 1. Explain the reasoning clearly and logically.
> 2. Present the relevant intermediate steps needed to understand the answer.
> 3. Do not introduce irrelevant information.
> 4. Clearly state the final answer at the end.
>
> Question: [question text]

The resulting response therefore contained both an explanation and a final answer. The two prompts share the same framing sentence and differ in what they require of the response: the standard prompt prohibits reasoning and extended explanation, while the CoT prompt requires reasoning and intermediate steps. Both prohibit irrelevant information.

The same questions and the same LLM were used in both conditions. The intended main difference between the conditions was the presence or absence of an explanatory reasoning component. However, the CoT responses were naturally longer and contained more information, meaning that response length and amount of information could not be completely separated from the explanation itself.

The prompts, model version, and relevant generation settings were documented to support reproducibility.

Figure 3.2 shows the two response formats side by side for one of the ten questions, to illustrate the difference participants were asked to evaluate.



Note. Both conditions were generated by the same model, using the same generation settings, and produced the same final answer. Only the presentation instruction differed. As discussed in Section 3.2.1, the CoT-style responses were consequently also longer and more detailed, so response length is not separated from the presence of an explanation.

Figure 3.2: The two response conditions compared, shown for one of the ten experimental questions.

3.4.4 Response Review and Validation

Before the responses were presented to participants, they were reviewed to check whether they were suitable for the experiment. The review focused on correctness, relevance, clarity, and consistency between the two response conditions.

For each of the ten questions, the final answer in the standard response was compared with the final answer in the corresponding CoT response. Where applicable, the answers were also checked against the expected solution to the question.

The main purpose of this review was to make sure that the two conditions gave the same intended final answer. This was important because the study was designed to compare participants' evaluations of responses with and without explanations rather than responses that produced different answers.

The responses were also checked for relevance and clarity. Responses containing inconsistent final answers or unclear or irrelevant content were not used in the participant evaluation.

Table 3.3 summarises the validation of the ten response pairs.

Table 3.3: Validation of standard and CoT responses

Question	Standard answer	CoT answer
Q1	€0.05	€0.05
Q2	No	No
Q3	60 minutes	60 minutes
Q4	312211	312211
Q5	€135	€135
Q6	Yes	Yes
Q7	No	No
Q8	Priority based on impact, dependencies and risk	Priority based on impact, depend
Q9	Cannot be determined	Cannot be determined
Q10	Evidence such as clinical trials and reliable sources	Evidence such as clinical trials ar

The review confirmed that the ten response pairs used in the study had consistent intended final answers. The CoT responses were generally longer because they included additional explanatory information. Therefore, although the final answers were kept consistent, the two conditions were not identical in terms of response length or amount of information. These differences are considered when discussing the limitations of the study.

3.5 Survey Design

The questionnaire was created using Google Forms. It contained questions about participants' background and experience with AI tools, followed by the evaluation of the generated responses.

Before starting the questionnaire, participants were provided with information about the study and gave their consent to participate.

For each response, participants rated their perceptions using a five-point Likert scale. The questions covered understanding, trust, confidence, transparency, and the clarity and usefulness of the explanation.

The six evaluation statements were written for this study rather than taken from an existing instrument. Validated scales for measuring trust in automation do exist, and Kohn et al. [30] catalogue the principal ones together with guidance on selecting among them; Sheridan [33] discusses the attributes such instruments attempt to capture, and Rodriguez Rodriguez et al. [34] review attempts to model trust formation formally. None was adopted here for two reasons. These instruments are formulated for settings in which an operator supervises an automated system over a period of time, whereas participants in this study rated twenty individual responses in a single session; and no validated scale was identified for the specific constructs of perceived transparency and explanation quality as applied to natural-language explanations.

The cost of that decision is that the six statements have no established measurement properties. Three of them, covering perceived transparency, explanation clarity and usefulness, and process understanding, are single items, so their internal consistency cannot be assessed at all. This is stated as a limitation in Section 5.10, and adopting a validated instrument is the first recommendation in Section 6.1.

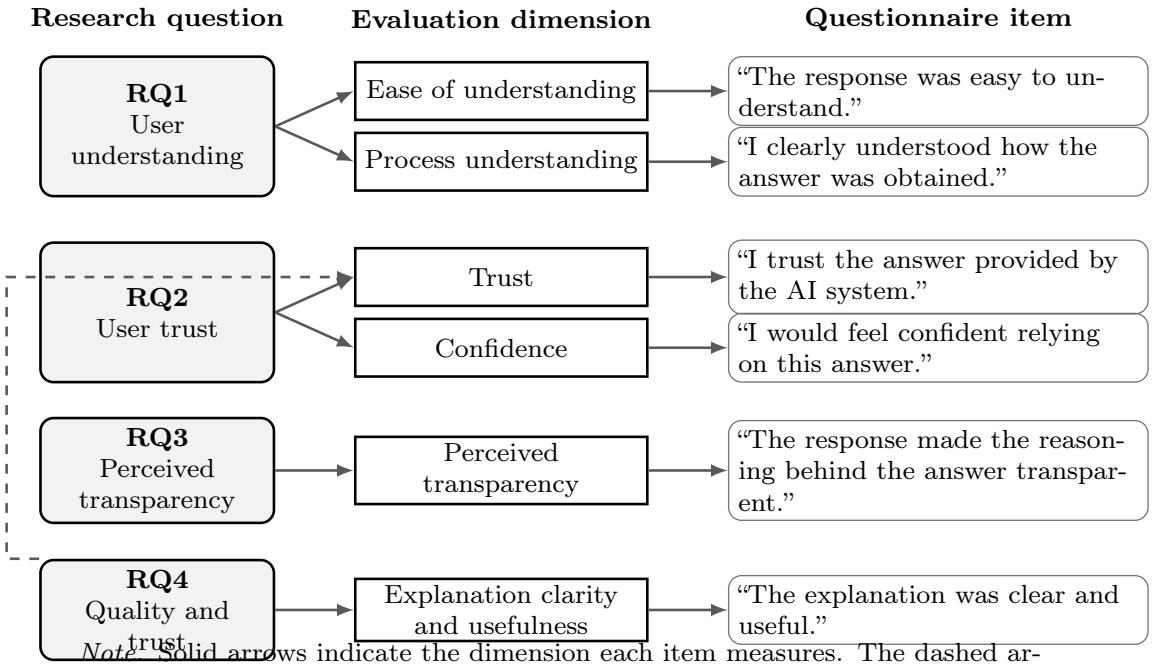
The Likert scale used in the questionnaire was:

- 1 – Strongly Disagree
- 2 – Disagree
- 3 – Neither Agree nor Disagree
- 4 – Agree
- 5 – Strongly Agree

The response evaluation statements were:

- The response was easy to understand.
- I clearly understood how the answer was obtained.
- I trust the answer provided by the AI system.
- I would feel confident relying on this answer.
- The response made the reasoning behind the answer transparent.
- The explanation was clear and useful.

Figure 3.3 shows how each questionnaire item maps onto an evaluation dimension and onto the research question it was included to address.



Note. Solid arrows indicate the dimension each item measures. The dashed arrow shows that RQ4 additionally uses the trust dimension, since it examines the association between perceived explanation quality and trust rather than a difference between conditions.

Figure 3.3: Mapping between the research questions, the six evaluation dimensions, and the questionnaire items used to measure them.

3.6 Participants

A total of 25 participants took part in the study. All participants had previous experience with AI-based tools or Large Language Models, which gave them some familiarity with evaluating AI-generated responses.

The participants came from both technical and non-technical academic or professional backgrounds. Background information was collected to describe the sample and included AI-tool usage frequency, self-reported AI experience, and academic or professional background. No personally identifiable information was collected.

Twenty participants (80%) reported using AI tools daily, while five participants (20%) reported using them several times per week. In terms of self-reported experience, 14 participants (56%) described their experience as advanced and 11 participants (44%) described it as intermediate.

The participants represented a range of academic and professional backgrounds. Ten participants (40%) reported a background in Data Science or Artificial Intelligence, five (20%) reported an Engineering background, four (16%) reported a Computer Science or Information Technology background, four (16%) reported another technical background, and two (8%) reported a non-technical background.

The sample therefore consisted mainly of frequent AI users with intermediate or advanced self-reported experience. This familiarity was useful when evaluating the

AI-generated responses. At the same time, previous experience with AI may have affected participants' expectations about explanations and should therefore be considered as a possible source of bias.

The final sample of 25 participants was within the planned target range of 20–30 participants. No prospective statistical power analysis was conducted before data collection. Therefore, the sample size should not be interpreted as demonstrating sufficient statistical power for every analysis, particularly the correlation analysis examining explanation quality and trust.

3.7 Experimental Procedure

The experiment was carried out in several stages.

First, the ten selected questions were used to generate two responses for each question using the same LLM. One response followed the standard prompt and the other followed the CoT prompt.

The generated responses were then reviewed and validated before they were included in the questionnaire.

Participants first completed the consent and background sections of the questionnaire. They then evaluated the generated responses under both conditions. For each response, they provided ratings related to understanding, trust, confidence, transparency, and explanation quality.

The two conditions were presented in a fixed order, as Response A followed by Response B, and participants were not told which block represented the standard condition and which the CoT condition.

After completing the response evaluations, the participants' ratings were recorded and exported for data preparation and statistical analysis.

3.8 Data Collection

The questionnaire was distributed online using Google Forms. Participants completed the questionnaire individually and evaluated the responses without interaction with other participants.

The collected dataset contained participant background information and Likert-scale ratings for both the standard and CoT response conditions. The questionnaire did not collect names, email addresses, or other information that directly identified participants.

3.9 Data Preparation

After the questionnaire responses were collected, the data were exported and prepared for analysis.

The following steps were carried out:

1. The dataset was checked for completeness.
2. Incomplete or invalid responses were checked.
3. Likert-scale responses were converted to numerical values from 1 to 5.
4. Responses were organised according to the standard and CoT conditions.
5. Participant-level scores were calculated for the relevant evaluation measures.
6. The final dataset was prepared for statistical analysis.

Anonymous participant identifiers were used during the analysis.

3.10 Data Analysis

The survey data were analysed quantitatively to compare participants' evaluations of the standard and CoT response conditions. Because every participant evaluated both conditions, the observations were paired.

First, descriptive statistics were calculated for the evaluation items. Mean scores and standard deviations were calculated for the six evaluation dimensions: ease of understanding, understanding how the answer was obtained, trust, confidence, perceived transparency, and explanation clarity and usefulness.

Overall participant-level scores were also calculated for the two response conditions. These scores were used to compare the general evaluation of CoT and standard responses.

The Wilcoxon signed-rank test was used to test whether there was a statistically significant difference between the overall participant-level scores for the two conditions. This test was selected because the study used paired observations and the ratings were collected using a five-point Likert scale.

A significance level of $\alpha = 0.05$ was used. A p-value below 0.05 was considered statistically significant.

For RQ4, Spearman's rank correlation coefficient (ρ) was used to examine the relationship between explanation quality and trust in the CoT condition. The analysis examined whether participants who gave higher ratings to the clarity and usefulness of the explanation also tended to give higher trust ratings.

The quantitative analysis therefore included:

- Mean and standard deviation for the evaluation measures.
- Wilcoxon signed-rank test for the paired comparison between CoT and standard responses.
- Spearman's rank correlation for the relationship between explanation quality and trust.

3.10.1 Data Quality and Reliability

Several steps were taken to maintain consistency during the study. The same LLM, model configuration, and questions were used for both response conditions. The generated responses were also reviewed before they were shown to participants.

The questionnaire data were checked for missing or invalid entries before the statistical analysis. The final dataset contained complete ratings from all 25 participants, with no missing values in the evaluation items.

Internal consistency was assessed only for groups of items that were intended to measure the same construct. The two understanding-related items were treated as one group, and the two trust-related items were treated as another group. Cronbach's alpha was used to assess the internal consistency of these grouped items. Perceived transparency and explanation clarity and usefulness were measured using single items and were therefore not assessed using Cronbach's alpha.

The within-participant design also provided a consistent basis for comparison because every participant evaluated both response conditions. This reduced the influence of differences between participants when comparing the two conditions.

The sample size and participant characteristics were considered when interpreting the findings. The results are therefore discussed primarily in relation to the participants included in this study and should not be assumed to represent all users of LLM-based systems.

Internal consistency was assessed for the two pairs of items that were intended to measure the same construct, namely the two understanding items and the two trust items. The coefficients are reported in Table 3.4 and they are low within condition.

Table 3.4: Cronbach's alpha for the paired items

Item pair	**CoT-style**	**Standard**
Understanding items (ease and process)	0.55	0.02
Trust items (trust and confidence)	0.85	0.49

Note. $n = 25$, computed on participant-level scores within each condition. Coefficients computed on the two conditions pooled are substantially higher, 0.94 and 0.97, but those values are inflated by the large between-condition difference and do not describe internal consistency within a condition, so they are not reported as reliability estimates.

Three qualifications apply to these coefficients. First, each construct is measured by only two items, and Cronbach's alpha is unstable and difficult to interpret for two-item scales; the values should not be read as they would be for a multi-item instrument. Second, the coefficients differ markedly between conditions, and the value of 0.02 for the understanding items in the standard condition indicates that the two items were effectively uncorrelated there. A plausible explanation is restricted variance: standard-condition ratings were concentrated near the lower end of the scale, with standard deviations of 0.34 and 0.22, and a correlation cannot be estimated reliably when variance is compressed. Third, perceived transparency and explanation clarity are single items, so no coefficient can be computed for them at all.

The conclusion to draw is that the instrument's internal consistency is not established. This is stated as a limitation in Section 5.10, and adopting a validated multi-item instrument is the first recommendation in Section 6.1.

3.11 Ethical Considerations

Participation in the study was voluntary and based on informed consent. Participants were informed about the purpose of the study, the evaluation procedure, and how the collected data would be used.

No personally identifiable information was collected through the questionnaire. The responses were treated confidentially and analysed in aggregated form. Participants were also informed that they could choose not to participate or withdraw from the study without negative consequences.

The study did not involve sensitive personal questions or activities expected to cause physical or psychological harm. Participants were only asked to evaluate AI-generated responses and provide their perceptions regarding understanding, trust, transparency, and explanation quality.

To reduce potential bias, participants were not explicitly informed which responses represented the standard condition and which represented the CoT condition. The presentation order, however, was the same for every participant, so condition and position are confounded (Section 5.9).

All collected data were stored securely and were used only for the purposes of this research. The results were reported in aggregated form so that individual participants could not be identified.

Chapter 4

Results and Analysis

This chapter presents the results of the user evaluation of Chain-of-Thought (CoT)-style and standard LLM responses. A total of 25 participants completed the questionnaire. Each participant evaluated both response conditions across ten reasoning-related questions. For each response, participants provided ratings on six five-point Likert-scale statements covering understanding, process understanding, trust, confidence, perceived transparency, and explanation clarity and usefulness.

One framing point applies throughout this chapter. All six evaluation dimensions are self-reported perception measures. Ratings of understanding record perceived understanding rather than verified comprehension; ratings of trust record self-reported willingness to rely on an answer rather than its reliability; and ratings of transparency record the extent to which participants felt able to see how an answer was reached, not access to the model's internal computation. Results are reported in those terms consistently below, and the distinction is maintained in Chapter 5.

The analysis was performed at the participant level. For each participant, the ratings for each evaluation dimension were averaged across the ten questions separately for the CoT-style and standard conditions. These participant-level scores were then used for the statistical comparisons. This approach ensured that the repeated ratings from the same participant were not treated as independent observations.

4.1 Participant Characteristics

A total of 25 participants completed the study. Twenty participants (80%) reported using AI tools daily, while five participants (20%) reported using AI tools several times per week.

Regarding AI experience, 14 participants (56%) described their experience as advanced and 11 participants (44%) as intermediate.

The participants also represented different academic and professional backgrounds. Ten participants (40%) reported a background in Data Science or Artificial Intelligence, five (20%) in Engineering, four (16%) in Computer Science or Information Technology, four (16%) in other technical fields, and two (8%) in non-technical fields.

Table 4.1: Participant characteristics

Characteristic	Number	Percentage
AI usage frequency		
Daily	20	80%
Several times a week	5	20%
AI experience		
Intermediate	11	44%
Advanced	14	56%
Academic/professional background		
Data Science / AI	10	40%
Engineering	5	20%
Computer Science / IT	4	16%
Other technical field	4	16%
Non-technical field	2	8%
Total participants	**25**	**100%**

4.1.1 Composition of the Sample

Three characteristics of the sample should be stated before the results are presented, because each bounds what the ratings can be taken to represent.

First, no demographic information beyond AI-tool usage, self-assessed AI experience and academic or professional background was collected. The questionnaire contained three background items, reproduced in Appendix A.1.2, and none of them recorded gender, age or country. The consequence is not that an effect of these characteristics goes unreported, since 25 participants could not support such an analysis in any case, but that a reader cannot judge how far the sample resembles the population of LLM users to which the findings might be extended. Collecting at least gender and age band would be straightforward in a replication and is recommended in Section 6.1.

Second, every participant was an existing user of AI tools, and 80 per cent reported daily use. The sample therefore consists of people already accustomed to reading model-generated text, who may have formed expectations about what such output looks like. Yang et al. [32] show experimentally that prior experience with an automated system affects reported trust independently of the system's reliability, so this characteristic is likely to have shaped the ratings rather than being incidental to them.

Third, 92 per cent of participants reported a technical background, and 40 per cent specifically in data science or artificial intelligence. Participants were therefore better placed than a general user to evaluate reasoning steps, and correspondingly less representative of users who encounter LLM output without that preparation.

The sample was therefore composed mainly of frequent AI users with intermediate or advanced self-reported experience. This familiarity with AI systems provided participants with prior experience for evaluating AI-generated responses. At the same time, previous experience may have influenced participants' expectations or preferences regarding explanatory responses and should therefore be considered when interpreting the results.

4.2 Reporting Conventions

The following conventions apply to every statistic reported in this chapter, and are stated here so that each figure can be read precisely.

Ratings were recorded on a five-point scale with the verbal anchors given in Section 3.5. All differences are expressed in scale points, so a difference of 1.66 means 1.66 points on a scale whose usable range is four points, from 1 to 5. Means and standard deviations are reported to two decimal places. Differences are computed from unrounded means and may therefore differ in the second decimal place from the subtraction of the rounded values displayed in the tables; where this occurs it is noted beneath the table concerned.

All inferential statistics were computed on participant-level scores, giving $n = 25$ independent observations per condition rather than the 3,000 individual ratings collected. Wilcoxon signed-rank tests were two-tailed with $\alpha = 0.05$, and the six comparisons across the evaluation dimensions were adjusted using the Holm procedure. Effect sizes are reported as $r = |Z|/\sqrt{N}$ and interpreted using the conventional thresholds, under which $r \geq 0.5$ constitutes a large effect. Correlations are Spearman's ρ, reported with the exact two-tailed p value.

4.3 Data Structure and Aggregation

Each participant evaluated ten questions under two response conditions: CoT-style and standard. Six ratings were collected for each response. Consequently, the questionnaire produced:

$$25 \times 10 \times 2 \times 6 = 3{,}000$$

individual ratings.

These ratings were repeated observations from the same participants and were therefore not treated as 3,000 independent observations.

The six evaluation statements measured:

1. ease of understanding;
2. understanding of how the answer was obtained;
3. trust in the AI response;
4. confidence in relying on the response;
5. perceived transparency; and
6. explanation clarity and usefulness.

For each participant and evaluation dimension, the ten question-level ratings were averaged separately for the CoT-style and standard conditions. This resulted in one participant-level score for each dimension and condition. These participant-level scores were used in the subsequent statistical analysis.

Descriptive statistics included means and standard deviations. Because the same participants evaluated both response conditions, differences between the conditions were examined using the Wilcoxon signed-rank test. This non-parametric paired test was selected because the observations were paired within participants and the questionnaire responses were measured using five-point Likert-scale items.

Effect size was expressed as

$$r = \frac{|Z|}{\sqrt{N}},$$

where Z is the standardized test statistic and N is the number of non-zero paired observations.

For RQ4, Spearman's rank correlation was used to examine the association between the rating of explanation clarity and usefulness and the corresponding trust rating. This analysis was performed at the participant level.

4.4 Descriptive Results

Table 4.2 presents the descriptive statistics for the six evaluation dimensions in both conditions, and Table 4.3 the corresponding test statistics.

Table 4.2: Comparison of CoT-style and standard responses

Dimension	**CoT-style**		**Standard**	
	M (SD)	Mdn [IQR]	*M* (SD)	Mdn [IQR]
Ease of understanding	4.19 (0.50)	4.10 [4.00, 4.40]	2.53 (0.34)	2.40 [2.30, 2.70]
Process understanding	4.23 (0.45)	4.30 [4.10, 4.40]	2.44 (0.22)	2.40 [2.30, 2.60]
Trust	4.11 (0.37)	4.20 [4.10, 4.30]	2.59 (0.26)	2.60 [2.40, 2.80]
Confidence	4.11 (0.40)	4.10 [4.00, 4.40]	2.51 (0.24)	2.50 [2.40, 2.60]
Perceived transparency	4.10 (0.37)	4.20 [3.90, 4.30]	2.51 (0.23)	2.50 [2.40, 2.60]
Explanation clarity and usefulness	4.01 (0.37)	4.10 [3.80, 4.30]	2.40 (0.19)	2.40 [2.30, 2.50]

Note. $n = 25$. Values in parentheses are standard deviations; values in brackets are the first and third quartiles. All statistics are computed on participant-level scores, each being the mean of that participant's ten ratings for the dimension and condition concerned. Medians and quartiles are reported alongside the means because the underlying responses are ordinal. Test statistics are given in Table 4.3.

Table 4.3 reports the corresponding Wilcoxon signed-rank tests.

The CoT-style condition received higher mean ratings than the standard condition on all six evaluation dimensions. The largest difference was observed for process understanding, where the mean rating was 4.23 for CoT-style responses and 2.44 for standard responses, giving a difference of 1.79 points.

The smallest difference was observed for trust, where CoT-style responses received a mean rating of 4.11 compared with 2.59 for standard responses, a difference of 1.52 points. It should be noted that the six differences fall within a narrow band, from 1.52 to 1.79 points, so trust represents the smallest of six large differences rather than a small difference in absolute terms. Section 4.5 considers why trust moved least.

Table 4.3: Wilcoxon signed-rank tests comparing the two conditions

Dimension	**Diff.**	W	Z	p	$p_{\mathbf{Holm}}$	r
Ease of understanding	1.66	0	−4.374	5.96×10^{-8}	3.58×10^{-7}	0.88
Process understanding	1.79	0	−4.373	5.96×10^{-8}	3.58×10^{-7}	0.88
Trust	1.52	1	−4.347	1.19×10^{-7}	3.58×10^{-7}	0.87
Confidence	1.60	0	−4.374	5.96×10^{-8}	3.58×10^{-7}	0.88
Perceived transparency	1.59	0	−4.376	5.96×10^{-8}	3.58×10^{-7}	0.88
Explanation clarity and usefulness	1.62	0	−4.375	5.96×10^{-8}	3.58×10^{-7}	0.88

Note. $n = 25$, two-tailed. *Diff.* is the difference in means in scale points, computed from unrounded values. W is the Wilcoxon signed-rank statistic and p its exact two-tailed probability; Z is the normal approximation without continuity correction, from which the effect size $r = |Z|/\sqrt{N}$ is derived. Holm-adjusted probabilities are given for the family of six comparisons. A value of $W = 0$ indicates that every participant rated the CoT-style condition higher on that dimension; $W = 1$ for trust reflects one participant who rated the standard condition higher, by 0.10 scale points. All six comparisons remain significant at $\alpha = 0.05$ after adjustment, and all six effect sizes exceed the conventional threshold of 0.5 for a large effect.

All six paired comparisons were statistically significant ($p < 0.001$), and the corresponding effect sizes were large. The consistent direction of the differences indicates that the higher ratings for CoT-style responses were observed across the participant group rather than being driven only by a small number of participants.

Figure 4.1 shows the distribution of the participant-level scores behind those summary statistics.

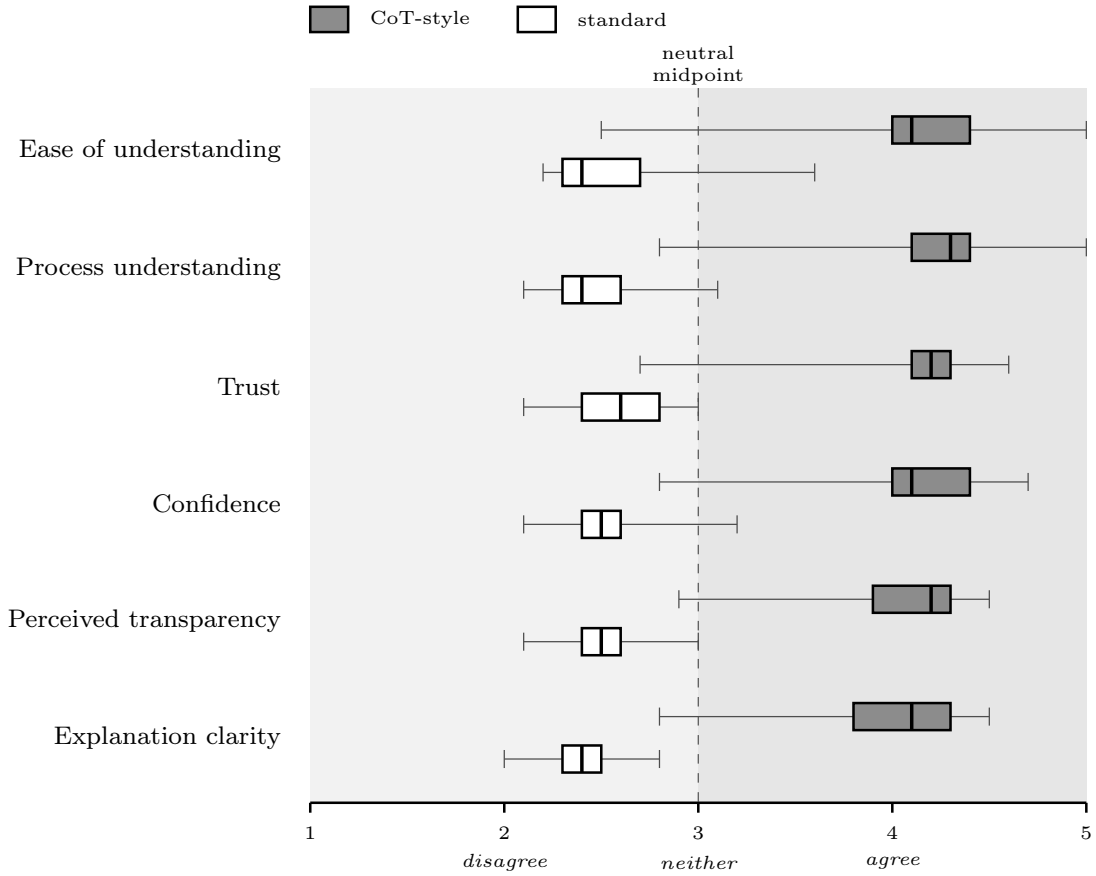
4.5 Interpretation of the Observed Differences

The preceding section reports the magnitude of the differences. This section states what those magnitudes mean on the response scale used, since a difference expressed in scale points is otherwise difficult to interpret.

4.5.1 The Two Conditions Fall in Different Response Categories

The five-point scale used in this study has defined verbal anchors, where 2 corresponds to *disagree*, 3 to *neither agree nor disagree*, and 4 to *agree*. Read against those anchors, the results have a straightforward interpretation: on every one of the six dimensions, the mean rating for standard responses falls between 2.40 and 2.59, and the mean rating for CoT-style responses falls between 4.01 and 4.23. The standard condition therefore sits in the *disagree* region of the scale throughout, and the CoT condition sits in the *agree* region throughout.

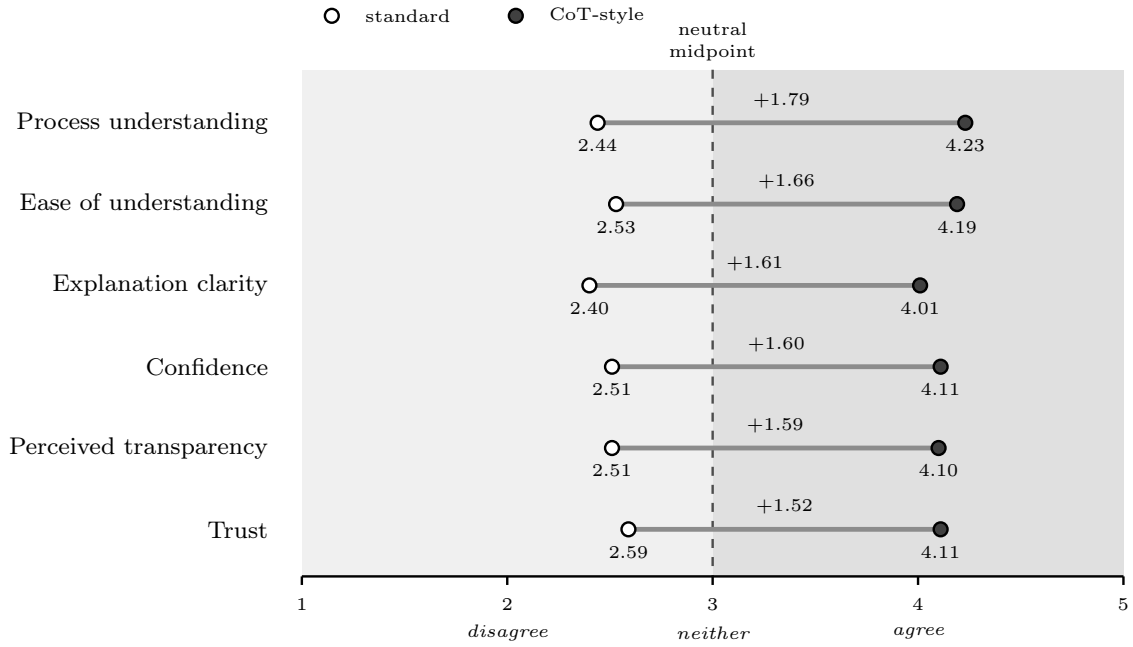
This is a stronger statement than the point differences alone convey. The two conditions did not merely differ in degree; on average, participants disagreed that standard responses were understandable, trustworthy or transparent, and agreed that CoT-style responses were. Every dimension crossed the neutral midpoint of the scale.



Note. Each box summarises the 25 participant-level scores for one dimension in one condition. The box spans the first to the third quartile, the heavy line is the median, and the whiskers extend to the minimum and maximum observed values rather than to a multiple of the interquartile range, so the full range of the data is visible. The dashed line marks the neutral midpoint of the scale.

Figure 4.1: Distribution of participant-level scores by dimension and condition ($n = 25$).

Figure 4.2 presents the same results against the verbal anchors of the response scale, which makes the position of each condition relative to the neutral midpoint visible.



Note. Mean participant-level ratings on the five-point scale ($n = 25$). Every standard-condition mean falls between 2.40 and 2.59, and every CoT-condition mean falls between 4.01 and 4.23. On all six dimensions the two conditions therefore fall on opposite sides of the neutral midpoint. Values beside each line are the mean differences in scale points.

Figure 4.2: Mean ratings for the two conditions positioned against the verbal anchors of the response scale.

4.5.2 Why Trust Showed the Smallest Difference

Table 4.2 shows that the difference for trust, 1.52 points, is the smallest of the six, and that process understanding, at 1.79 points, is the largest. The reason is visible in the baseline rather than in the CoT condition. Trust recorded the *highest* mean of any dimension in the standard condition, at 2.59, whereas process understanding recorded the lowest, at 2.44. Participants were somewhat more willing to trust an unexplained answer than to state that they understood how it had been obtained, so trust began from a more generous position and consequently had less distance to travel.

The CoT means, by contrast, are tightly clustered between 4.01 and 4.23. The variation in the size of the differences is therefore driven mainly by differences in how the standard responses were rated, not by variation in how the CoT responses were received.

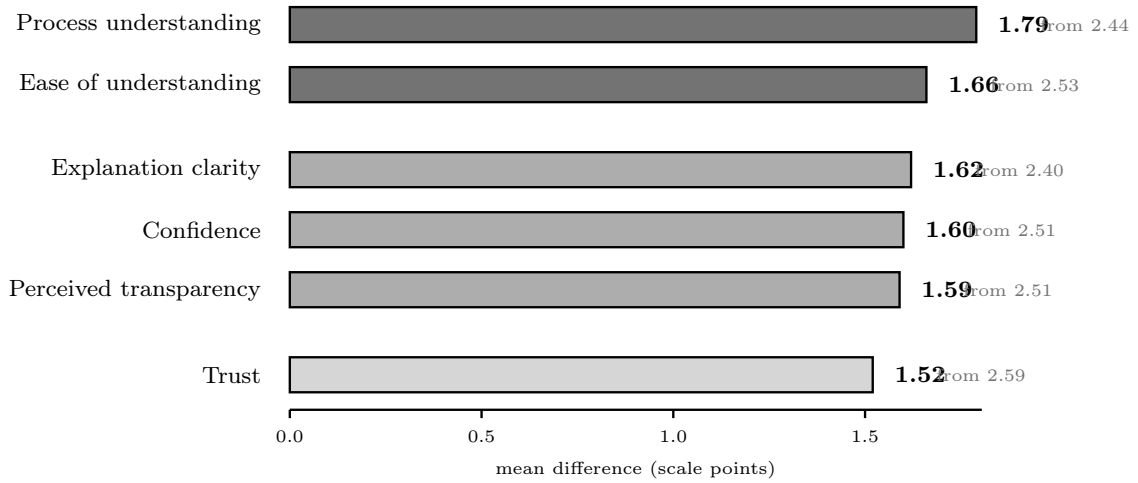
4.5.3 The Ordering of the Dimensions

Ranked by difference, the dimensions fall into three groups. The two understanding dimensions are largest, at 1.79 and 1.66 points. The explanation clarity, confidence and transparency dimensions are intermediate, at 1.62, 1.60 and 1.59 points. Trust is smallest, at 1.52 points.

This ordering is consistent across the two ends of the scale and is informative for the interpretation offered in Chapter 5. Participants conceded understanding and transparency more readily than they conceded trust. Seeing the reasoning moved their sense of comprehension further than it moved their willingness to rely on the answer. Section 5.7 returns to this point, since it bears directly on the RQ4 result.

One further observation concerns explanation clarity and usefulness, which received the lowest CoT-condition mean of the six dimensions at 4.01. The dimension that asked most directly about the explanation itself was rated slightly lower than the dimensions asking about its effects. This is relevant to RQ4, where the association between perceived explanation quality and trust did not reach significance.

Figure 4.3 presents the six differences in rank order, together with the standard-condition mean from which each one starts.



Note. Shading indicates the three groups described in the text: the two understanding dimensions, the three intermediate dimensions, and trust. The baseline given inside each bar is the standard-condition mean for that dimension. Trust shows the smallest difference and also the highest baseline, whereas process understanding shows the largest difference and the lowest baseline.

Figure 4.3: Mean differences between the two conditions in rank order, with the standard-condition baseline for each dimension.

4.5.4 Practical Implications for Users

Three implications follow for the design of user-facing LLM systems.

First, presentation format alone moved every measured perception across the midpoint of the scale, without any change to the model, the questions or the final answers. The way a response is presented is therefore not a cosmetic property of an LLM interface but a determinant of how users judge its output.

Second, the effect on trust is the one that warrants caution rather than celebration. Participants reported trust of 4.11 for responses whose correctness they were not told and mostly could not verify. A presentation change raised reported trust by more than a point and a half on a five-point scale while the underlying reliability of the answers was unchanged by construction, since both conditions produced the same final answers. For system designers this identifies a mechanism by which explanation can encourage reliance that is not grounded in accuracy.

Third, the gap between the understanding dimensions and the trust dimension suggests that explanation and reassurance are separable design goals. An interface that makes reasoning visible can be expected to improve users' sense of comprehension substantially, but should not be assumed thereby to have produced appropriately calibrated trust. Section 5.11 develops this into concrete design recommendations.

4.6 Overall Response Evaluation

For descriptive purposes, an exploratory overall evaluation index was calculated for each participant and condition by averaging the six evaluation dimensions across the ten questions. This index is used only to summarise the overall response ratings and is not treated as a validated multi-item psychological scale.

The mean overall score was 4.12 for CoT-style responses (SD = 0.34, Mdn = 4.20, IQR = $[4.02, 4.33]$) and 2.50 for standard responses (SD = 0.16, Mdn = 2.48, IQR = $[2.40, 2.53]$). Computed from unrounded values the means are 4.1247 and 2.4960, giving a difference of 1.6287 points. The Wilcoxon signed-rank test on the index gives $W = 0$, $Z = -4.373$, exact $p = 5.96 \times 10^{-8}$ and $r = 0.88$.

Table 4.4: Exploratory overall evaluation of CoT-style and standard responses

Condition	**Mean**	**Median**
CoT-style response	4.12	4.00
Standard response	2.50	2.00

Note. The difference was calculated using unrounded mean values, so it may not equal the subtraction of the displayed rounded means.

All 25 participants had a higher exploratory overall evaluation score for the CoT-style condition than for the standard condition. The participant-level differences ranged from 0.2833 to 2.0667 points.

The same unanimity holds for five of the six individual dimensions. The exception is trust, where 24 of the 25 participants rated the CoT-style condition higher and one participant rated the standard condition higher by 0.10 scale points. This is the reason the trust comparison yields $W = 1$ rather than $W = 0$ in Table 4.3, and it is a further respect in which trust behaves slightly differently from the other dimensions, as discussed in Section 4.5.

4.7 RQ1: Effect of CoT Explanations on User Understanding

RQ1 examined how Chain-of-Thought explanations influence user understanding of LLM-generated responses. In this study, understanding was measured using two questionnaire items: ease of understanding and understanding of how the answer was obtained.

Table 4.5: User understanding under CoT-style and standard conditions

Measure	**CoT**	**Standard**	**Difference**	p	r
Easy to understand	4.19 (0.50)	2.53 (0.34)	1.66	< 0.001	0.88
Process understanding	4.23 (0.45)	2.44 (0.22)	1.79	< 0.001	0.87

Participants reported higher perceived understanding for CoT-style responses on both measures. Both paired comparisons were statistically significant, with large effect sizes.

4.7.1 Answer to RQ1

The results show that participants gave higher self-reported perceived understanding ratings to the CoT-style responses than to the standard responses. This difference was observed for both ease of understanding and understanding of how the answer was obtained.

However, these results should not be interpreted as evidence that Chain-of-Thought reasoning alone caused the difference. The CoT-style responses were generally more detailed and longer than the standard responses. Therefore, the higher ratings may also reflect differences in response length, information content, structure, or explanation presence.

4.8 RQ2: Effect of CoT Explanations on User Trust

RQ2 examined whether participants reported different levels of trust between the two response conditions.

The mean trust rating was 4.11 (SD = 0.37) for CoT-style responses and 2.59 (SD = 0.26) for standard responses. The mean difference was 1.52 points.

The Wilcoxon signed-rank test indicated a statistically significant difference between the two conditions ($p < 0.001$), with a large effect size ($r = 0.87$).

4.8.1 Answer to RQ2

Participants reported higher trust ratings for CoT-style responses than for standard responses. Within this sample, the presence of the more detailed step-by-step response format was therefore associated with higher self-reported trust.

The result should, however, be interpreted with caution because the two conditions differed in response length and level of detail. The analysis does not establish

that Chain-of-Thought reasoning itself was responsible for the observed difference in trust.

4.9 RQ3: Effect of CoT Explanations on Perceived Transparency

RQ3 examined participants' perceived transparency of the two response conditions.

The mean transparency rating was 4.10 (SD = 0.37) for CoT-style responses and 2.51 (SD = 0.23) for standard responses. The mean difference was 1.59 points.

The Wilcoxon signed-rank test indicated a statistically significant difference between the two conditions ($p < 0.001$), with a large effect size ($r = 0.88$).

It should be noted that the transparency measure captures participants' perceptions of transparency. It does not establish that the explanation provides direct access to the actual internal reasoning process of the LLM.

4.9.1 Answer to RQ3

Participants reported higher perceived transparency for responses that included the CoT-style step-by-step explanation. Thus, the CoT-style condition was associated with higher perceived transparency in this study.

This finding should not be interpreted as evidence that the explanations represent the model's actual internal reasoning. It reflects participants' perceptions of the information presented in the response.

4.10 RQ4: Relationship Between Explanation Quality and User Trust

RQ4 examined the relationship between perceived explanation quality and user trust. In the questionnaire, explanation quality was represented by the statement "The explanation was clear and useful." Therefore, the measure is referred to here as perceived explanation clarity and usefulness rather than as a validated multi-item measure of explanation quality.

For the CoT-style responses, the Spearman rank correlation between perceived explanation clarity and usefulness and trust was positive ($\rho = 0.37$, $p = 0.067$, $n = 25$). The association did not reach statistical significance at the 0.05 level.

For comparison, the standard condition showed a weaker positive association ($\rho = 0.21$, $p = 0.315$).

Table 4.6: Spearman correlation between explanation clarity and usefulness and trust

Condition	**Spearman's ρ**	**p-value**
CoT-style	0.37	0.067
Standard	0.21	0.315

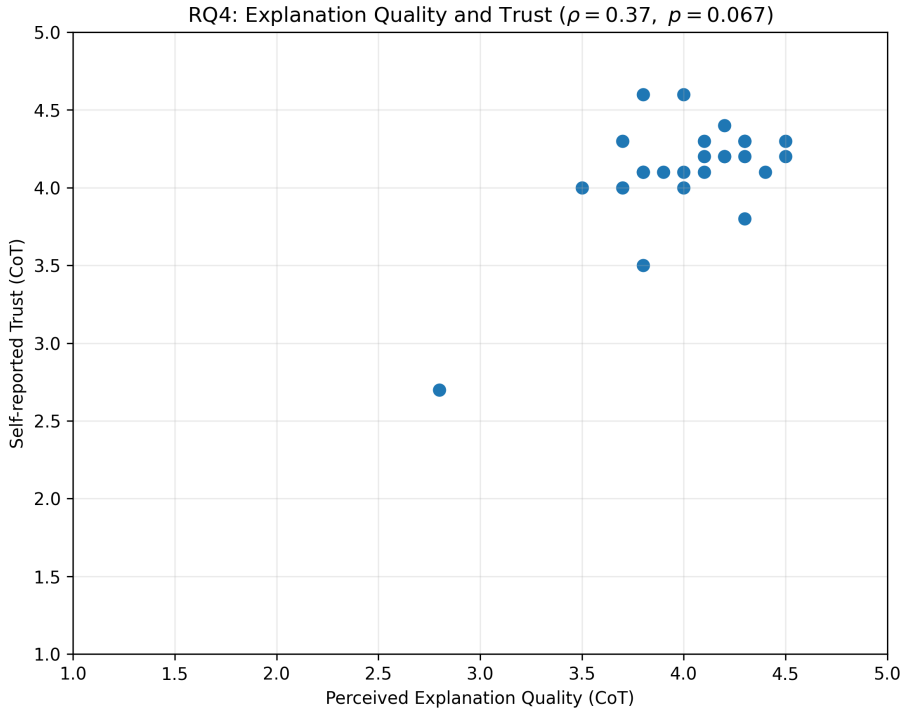


Figure 4.4: Relationship between perceived explanation clarity and usefulness and trust for CoT-style responses ($n = 25$).

The positive correlation in the CoT-style condition indicates that participants who gave higher ratings for explanation clarity and usefulness also tended to report higher trust ratings. However, because the correlation was not statistically significant, the study does not provide sufficient evidence to conclude that perceived explanation quality and trust were significantly associated in the sample.

4.10.1 Answer to RQ4

The results provide limited evidence of a positive association between perceived explanation clarity and usefulness and trust for CoT-style responses. However, the association was not statistically significant ($\rho = 0.37$, $p = 0.067$). Therefore, H4 is not supported by the statistical analysis.

4.11 Summary of Results

Table 4.7 summarises the main findings in relation to the four research questions.

Overall, participants rated the CoT-style responses higher than the standard responses across all six measured dimensions. The differences were statistically significant and were associated with large effect sizes.

These results describe differences in participants' self-reported perceptions. Because the CoT-style responses were generally longer and more detailed than the standard responses, the findings do not isolate an effect attributable solely to Chain-of-Thought

Table 4.7: Summary of findings in relation to the research questions

Research Question	**Finding**
RQ1: Effect on user understanding	CoT-style responses received higher self-reported perceived understanding ratings than standard responses for both understanding measures.
RQ2: Effect on user trust	CoT-style responses received higher self-reported trust ratings than standard responses, with a mean difference of 1.52 points.
RQ3: Effect on perceived transparency	CoT-style responses received higher perceived transparency ratings than standard responses, with a mean difference of 1.59 points.
RQ4: Relationship between explanation quality and trust	A positive association was observed for CoT-style responses ($\rho = 0.37$), but it was not statistically significant ($p = 0.067$).

reasoning.

The relationship between perceived explanation clarity and usefulness and trust was positive but did not reach statistical significance. The interpretation of these findings, their relationship to previous research, and the limitations of the study are discussed in the following chapter.

Chapter 5

Discussion

This chapter interprets the findings presented in the previous chapter and relates them to the research questions and existing literature. The study examined how participants evaluated standard and Chain-of-Thought (CoT)-style responses from a Large Language Model (LLM), with particular attention to perceived understanding, self-reported trust, confidence, perceived transparency, and explanation clarity and usefulness. It also examined whether perceived explanation quality was associated with user trust.

The results showed a consistent difference between the two experimental conditions. Participants gave higher ratings to CoT-style responses than to standard responses across understanding, trust, confidence, transparency, and explanation clarity and usefulness. The largest descriptive difference was observed for participants' understanding of how the answer was obtained. However, the relationship between perceived explanation quality and trust was positive but did not reach statistical significance.

These findings should be interpreted as evidence about participants' perceptions of the two response conditions rather than as evidence of a causal effect attributable solely to Chain-of-Thought reasoning. The standard responses were designed to be concise, whereas the CoT-style responses were longer and provided more detailed reasoning information. Consequently, differences in response length, information content, presentation format, and perceived effort may also have contributed to the observed results. In addition, the explanations presented to participants should not be interpreted as direct access to the actual internal reasoning process of the LLM.

5.1 Overview of the Main Findings

The findings indicate a substantial difference in how participants evaluated the two response conditions. Across all six evaluation dimensions, CoT-style responses received higher ratings than standard responses. The largest descriptive difference was observed for participants' understanding of how the answer was obtained. Participants also reported higher levels of trust, confidence, perceived transparency, and explanation clarity and usefulness for the CoT-style responses.

The statistical analysis supported the consistency of these differences. The within-participant comparisons showed statistically significant differences between the two conditions across all six evaluation dimensions. The effect sizes were large, indicating that the observed differences were substantial within the sample.

These findings are broadly consistent with research on explainable and human-centred AI, where explanations have been associated with improved understanding and increased trust. However, the present findings should be interpreted specifically as differences in participants' evaluations of standard concise responses and CoT-style step-by-step responses. The study does not isolate the contribution of CoT reasoning from other differences between the response formats.

5.1.1 Consistency of the Observed Response Pattern

An unusually consistent pattern was observed in the participant responses. All 25 participants provided a higher overall evaluation for the CoT condition than for the standard condition. In addition, the differences between the two conditions were relatively similar across the six evaluation dimensions.

This consistency provides strong descriptive evidence of a systematic difference in participants' evaluations within the study sample. The within-participant design is useful in this respect because each participant evaluated both response conditions, reducing the influence of differences between individual participants.

However, the uniformity of the responses also requires cautious interpretation. The present study cannot determine whether the observed pattern was attributable entirely to the explanatory content of the CoT responses. Other characteristics of the experimental materials, such as response length, amount of information, presentation format, level of detail, or participants' expectations about explanatory responses, may also have contributed to the pattern.

Therefore, the consistency of the findings should be considered a feature of the observed dataset rather than evidence that CoT-style explanations would necessarily produce the same results for all users, models, or contexts.

5.2 Discussion of RQ1: User Understanding

The first research question examined how CoT explanations influence users' understanding of LLM-generated responses. The results show that participants rated the CoT-style responses as easier to understand and as providing greater insight into how the answer was obtained. The largest descriptive difference among the six evaluation dimensions was observed for the statement concerning understanding how the answer was obtained.

This finding suggests that participants perceived the additional reasoning information as useful for interpreting the relationship between the question and the final answer. Rather than receiving only a final conclusion, participants were presented with intermediate explanatory information that could help them follow the steps leading to the response.

This finding is consistent with the original motivation behind Chain-of-Thought prompting. Wei et al. [2] demonstrated that generating intermediate reasoning steps can improve LLM performance on complex reasoning tasks, including arithmetic, commonsense, and symbolic reasoning. However, the present study examines a different aspect of CoT. Rather than measuring model accuracy or reasoning performance, it examines how users evaluate the resulting responses and explanations.

The findings therefore provide a user-centred perspective on the use of reasoning explanations. When explanatory information is presented in a form that participants find understandable, it may provide additional information for interpreting an AI-generated answer.

At the same time, the results should not be interpreted as proof that participants obtained a correct understanding of the LLM's actual reasoning process. The measure used in this study captured participants' perceived understanding rather than objective comprehension or verification of the model's internal reasoning.

This distinction is important because previous research has shown that Chain-of-Thought explanations can sometimes be unfaithful to the factors that actually produced a model's answer. Turpin et al. [26] found that language models can generate plausible reasoning explanations that rationalise an answer without accurately representing the model's underlying reasoning process.

Therefore, the higher understanding ratings observed in this study should be interpreted as higher perceived understanding of the presented response, rather than evidence that the explanation faithfully represented the internal computational process of the LLM.

5.3 Discussion of RQ2: User Trust

The second research question investigated how CoT explanations influence users' trust in LLM-generated responses. Participants reported substantially higher trust ratings for the CoT-style responses than for the standard responses. They also reported higher confidence in relying on the CoT-style responses.

One possible explanation is that reasoning explanations provide users with additional information that can make an otherwise opaque response appear more understandable and justifiable. When users can follow an explanation, they may feel better able to evaluate the response rather than receiving only an unsupported conclusion.

This interpretation is consistent with research on transparency in human–AI interaction. Previous experimental research has found that transparency can reduce uncertainty and increase trust in AI systems. Similarly, research examining transparency and explainability has reported positive relationships with trust, confidence, perceived reliability, and ease of understanding.

However, the higher trust ratings observed in the present study should not be interpreted as evidence that the CoT-style responses were objectively more reliable or correct. Trust and reliability are related but distinct concepts. A response may appear more convincing because it contains a coherent and detailed explanation, even when the underlying answer or reasoning is incorrect.

This issue is particularly relevant for LLMs because fluent explanations can create an impression of competence. Turpin et al. [26] demonstrated that CoT explanations may sometimes rationalise answers rather than faithfully represent the factors responsible for the model output. Consequently, the higher trust ratings observed in the present study should be understood as higher self-reported or perceived trust, rather than evidence that CoT-style responses necessarily produce more trustworthy AI outputs.

The finding also highlights an important consideration for the design of explana-

tory AI systems. If explanations increase users' confidence without providing reliable evidence about the correctness of the answer, there is a possibility that explanations could encourage inappropriate reliance. Therefore, perceived trust should be considered together with response correctness, explanation faithfulness, and appropriate communication of uncertainty.

5.4 Discussion of RQ3: Perceived Transparency

The third research question examined how CoT explanations influence users' perceived transparency of LLM responses. The results showed substantially higher perceived transparency ratings for the CoT-style responses than for the standard responses.

This finding is consistent with the broader explainable AI literature, where explanations are commonly considered a mechanism for making otherwise opaque AI systems more understandable to users. Explanatory information can provide users with additional information about how an output was produced and can consequently reduce uncertainty about the response.

The present study supports this user-centred interpretation of transparency. Participants did not have access to the underlying model or its computational process. Instead, transparency was assessed through their perception of the information presented in the response. The higher transparency ratings for CoT-style responses therefore indicate that participants perceived the reasoning explanation as providing greater visibility into the process described by the response.

However, perceived transparency should be distinguished from technical transparency and explanation faithfulness. A textual explanation may give a user the impression that the model's reasoning is visible without necessarily providing a faithful representation of the internal computational process.

This distinction is particularly important for interpreting the results of the present study. The study measured whether participants perceived the responses as transparent; it did not directly evaluate whether the explanations faithfully represented the internal operations of the LLM.

Therefore, the results should not be interpreted as demonstrating that CoT makes an LLM intrinsically or technically transparent. Rather, they show that participants reported higher perceived transparency when evaluating the CoT-style responses used in this study.

5.5 Discussion of RQ4: Relationship Between Explanation Quality and Trust

The fourth research question examined what relationship exists between perceived explanation quality and user trust. In the CoT condition, a positive association was observed between participants' ratings of explanation clarity and usefulness and their trust ratings. The Spearman correlation was $\rho = 0.37$ with $p = 0.067$ for $n = 25$ participants. Although the direction of the association was positive, it did not reach the conventional level of statistical significance.

This finding is important because it suggests that explanation quality and trust may be related but should not be treated as interchangeable constructs. Participants may consider an explanation clear and useful without necessarily reporting a correspondingly high level of trust in the underlying AI system.

One possible explanation is that trust in an LLM depends on factors beyond the quality of the explanation itself. Participants may also consider the apparent correctness of the answer, the type and difficulty of the question, their previous experience with AI, and their expectations about the reliability of the system.

The non-significant result is also a question of statistical power, and the two analyses in this thesis are not comparable in that respect. The within-participant comparison tests a large difference between two conditions and was well powered to detect it. The correlation analysis asks a different question, namely whether participants who perceived the explanation as clearer also reported higher trust, and an association of the observed magnitude requires a substantially larger sample. For a correlation of $\rho = 0.37$, approximately 55 participants would be needed to reach conventional power of 0.80 at $\alpha = 0.05$, rather than the 25 available here. The present result should therefore be read as insufficient evidence rather than as evidence of no association, and no prospective power analysis was conducted that would have identified this in advance.

A second and more interesting explanation lies in the distribution of the ratings themselves. Both variables entering the correlation were rated highly and with little spread in the CoT condition: explanation clarity and usefulness had a mean of 4.01 with a standard deviation of 0.37, and trust a mean of 4.11 with a standard deviation of 0.37 (Table 4.2). A correlation coefficient is attenuated when the variance of either variable is restricted, and here the variance of both was compressed by the consistency of the participants' responses. The very uniformity that makes the condition comparison convincing therefore works against detecting an association within the CoT condition. This is a design consequence rather than an incidental one: a study seeking to estimate the explanation-quality to trust relationship would need stimulus material that deliberately varies in explanation quality, rather than material in which the explanations are uniformly adequate.

Third, the result is not theoretically anomalous, and relating it to the literature makes it less surprising than the hypothesis implied. Turpin et al. [26] show that a CoT explanation may be plausible and clear while failing to represent the factors that actually produced the answer. If explanation clarity and answer reliability can come apart in the system, there is no reason to expect a reader's judgement of clarity to track their trust tightly, and some reason to expect the two to dissociate. The XAI literature [16,25] likewise treats explanation quality and warranted reliance as related but distinct evaluation targets rather than as a single construct. Read alongside that work, H4 embodied an assumption of proportionality between explanation quality and trust that the present data do not support and that existing theory does not require.

A final consideration concerns the measure. Explanation quality was captured by a single item, so its reliability could not be assessed and any measurement error attenuates the correlation further. A multi-item scale covering completeness, coherence and sufficiency, rather than clarity and usefulness alone, would provide a more precise estimate of the association.

Therefore, RQ4 cannot be answered affirmatively based on the present statistical evidence. The positive correlation provides an indication that perceived explanation quality may be associated with trust, but the evidence from this study is insufficient to establish a statistically significant relationship.

It is also important to note that the measure used for explanation quality was based on participants' ratings of whether the explanation was clear and useful. It should therefore be interpreted as perceived explanation quality rather than as an objective or validated measure of explanation quality.

5.6 Relation to Previous Research

This section states explicitly where the present findings agree with earlier work and where they diverge from it, since the two carry different weight for the contribution claimed in this thesis.

5.6.1 Findings That Confirm Previous Research

The direction of the effect on understanding is consistent with the premise of the CoT literature. Wei et al. [2] and Kojima et al. [3] report that making reasoning explicit improves model performance on reasoning tasks; the present study finds that making the same reasoning visible improves readers' reported understanding, with the largest of the six differences occurring on the item asking specifically about understanding how the answer was obtained. The two results concern different subjects, the model and the reader, but they point the same way.

The finding is also consistent with the central claim of the XAI literature [16, 20, 25], namely that exposing some account of how a system reached its output improves how users understand and evaluate it. The present study extends that claim from feature-based explanations of classical models to natural-language reasoning produced by an LLM, and finds it holds for perceived understanding and perceived transparency.

The caution about faithfulness is confirmed rather than challenged. Turpin et al. [26] argue that CoT explanations need not represent the process that produced the answer. Nothing in the present results contradicts this, and the transparency result gives it empirical weight from the user side: participants rated transparency at 4.10 on the basis of the presented text alone, having no access to the model's computation. High perceived transparency was obtained without any evidence of faithful transparency, which is precisely the dissociation Turpin et al. describe.

5.6.2 Findings That Diverge From Expectation

Two results depart from what the literature would predict.

The first is the RQ4 result. The implicit expectation in much writing on explainability is that better explanation produces greater warranted trust, and H4 was formulated on that basis. The observed association was positive but not significant ($\rho = 0.37$, $p = 0.067$), and the standard condition showed a weaker association still ($\rho = 0.21$, $p = 0.315$). As Section 5.5 argues, this is partly a matter of power and

restricted variance, but it is also a substantive divergence: explanation quality and trust behaved as related but separable constructs rather than as proportional ones. That result sits closer to Turpin et al. [26] than to the XAI literature's working assumption.

The second is the relative ordering of the dimensions. If CoT operated primarily by increasing trust in the system, trust would be expected to show the largest movement. It showed the smallest, at 1.52 points, while process understanding showed the largest, at 1.79 points. Participants conceded comprehension more readily than they conceded reliance. This pattern is not reported in the CoT literature, which does not measure these constructs separately, and it suggests that the mechanism by which CoT affects users is better described as improving apparent intelligibility than as improving credibility.

5.6.3 Where the Present Study Cannot Be Compared

One limitation of this comparison should be stated. Because the CoT responses in this study were also longer and more detailed, the present findings cannot be compared directly with studies that isolate the reasoning component. Wang et al. [17] show that the relevance and ordering of reasoning steps matter more than their validity for model accuracy; whether a comparable dissociation holds for human readers cannot be determined from the present design, which varied explanation presence and response length together. Studies using length-matched materials would be required, and this is the first item in Section 6.1.

5.7 Cross-RQ Discussion

The findings across the four research questions suggest that perceived understanding, perceived transparency, and self-reported trust are related aspects of users' evaluations of LLM responses, but they should not be considered identical constructs.

The strongest descriptive difference concerned participants' understanding of how the answer was obtained. Perceived transparency and self-reported trust were also higher for the CoT-style responses. At the same time, the correlation analysis did not establish a statistically significant relationship between perceived explanation quality and trust.

This pattern suggests that the relationship between explanations and trust may be more complex than a simple assumption that better explanations automatically result in higher trust. Explanations may provide users with additional information that supports understanding and perceived transparency, while trust may additionally depend on factors such as perceived correctness, task characteristics, previous experience with AI, and expectations about system reliability.

The findings therefore support a distinction between making an AI response appear more understandable and making the AI system objectively more reliable. A response can be easier to follow and appear more transparent without the explanation necessarily being faithful to the model's actual reasoning process.

For this reason, the relationships observed in this study should be understood as associations and differences in participant perceptions rather than as evidence of a

causal pathway between CoT explanations, understanding, transparency, and trust.

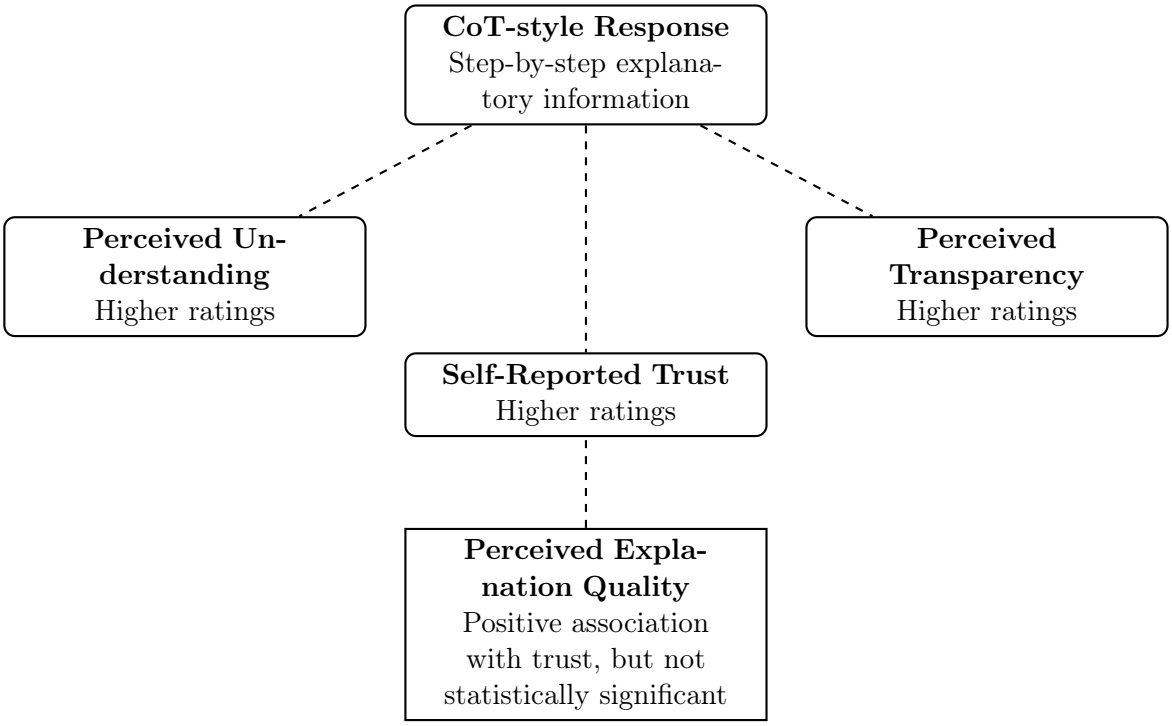


Figure 5.1: Non-causal summary of the main observed differences and association identified in the study. Dashed lines indicate observed relationships and should not be interpreted as causal pathways.

5.8 Benefits and Limitations of CoT-style Explanations

The findings indicate several potential benefits associated with the CoT-style responses used in this study. First, participants rated these responses higher in perceived understanding. Second, participants reported higher perceived transparency. Third, the CoT-style condition received higher ratings for self-reported trust and confidence.

These findings suggest that explanatory information can be valuable from a user-centred perspective. In particular, step-by-step information may help users understand how an answer is presented and may make the response appear more transparent and easier to evaluate.

However, the findings also highlight important limitations of using CoT-style explanations as a general solution for trustworthy AI. The experimental conditions differed in more than the presence of an explanation. The CoT-style responses were generally longer and more detailed than the standard responses. Therefore, the observed differences cannot be attributed solely to the use of Chain-of-Thought reasoning.

Response length, amount of information, formatting, level of detail, and participants' expectations may all have contributed to the observed differences. Future

studies should therefore consider designs that control these characteristics more carefully if the specific contribution of CoT explanations is to be isolated.

A further limitation concerns explanation faithfulness. A detailed explanation may increase perceived transparency without guaranteeing that the explanation faithfully represents the model's internal reasoning. In addition, an explanation may increase users' confidence even when the underlying answer is incorrect.

Consequently, explanation quality should not be evaluated only by whether users prefer or understand an explanation. Future evaluations should also consider whether explanations are accurate, relevant, faithful, and appropriately calibrated to the uncertainty of the model.

5.9 Order Effects

One feature of the procedure requires separate discussion because it was not recognised in the original design. The questionnaire presented the two conditions in two labelled blocks, Response A and then Response B, in the same order for every participant. Condition is therefore perfectly confounded with position. The response export establishes which block held which condition: block A contained the CoT-style responses and block B the standard responses, for every participant. The CoT-style condition was thus always evaluated first, and the condition that received the higher ratings was also the one seen first. This is the direction of confounding that matters, because any decline in attention across the 120 judgements would depress the later block and so inflate the apparent difference.

Three mechanisms could contribute to the observed difference as a result. Participants rated twenty responses on six statements each, a total of 120 judgements, so attention may have declined across the session and systematically disadvantaged the second block. Having already read ten explanatory responses, participants also had a reference point on reaching the second block that they lacked at the start, which may have made the terser standard responses appear poorer by contrast than they would have in isolation. And a participant who inferred the purpose of the study partway through may have responded differently in the second block from the first.

The size and consistency of the observed difference make it unlikely that order accounts for the whole effect: the difference exceeds 1.5 points on every dimension and all 25 participants rated the same condition higher. An order effect of that magnitude and uniformity would be unusual. The point is nevertheless that the design cannot separate the two, and a counterbalanced design, in which half the participants receive the CoT block first, is required to do so. This is recorded as the second recommendation in Section 6.1.

5.10 Limitations of the Trust Findings

The trust results require more qualification than the other three dimensions, because trust is the construct this study measures least directly and the one whose interpretation is most easily overstated. Five limitations apply.

Trust was self-reported, not observed. Participants stated agreement with the proposition that they trusted an answer. They were not asked to act on any

answer, to stake anything on it, or to choose between relying on the model and verifying the result themselves. Reported trust and demonstrated reliance are distinct, and the literature on trust in automation treats behavioural reliance as the outcome of interest rather than the stated attitude [30]. Nothing in these data shows what participants would have done with a CoT-style answer in a task that mattered.

No validated instrument was used. As stated in Section 3.5, the trust items were written for this study. Trust was captured by two statements, one on trust in the answer and one on confidence in relying on it, and neither has established measurement properties. A validated scale would permit comparison with other studies; the present figures can only be compared within this thesis.

Trust could not be calibrated against correctness. Participants were not told whether any answer was right, and for most of the ten questions they could not have verified it unaided. Reported trust of 4.11 in the CoT condition therefore records a judgement formed from the presentation of the response, not an assessment of its reliability. Because both conditions produced the same final answers by construction, the underlying accuracy was identical across conditions while reported trust differed by 1.52 points. Whether the higher figure represents better calibrated trust or simply more trust cannot be determined here. Lee and See's distinction between appropriate and maximal reliance [31] is precisely what this design cannot speak to, and Cabitza et al. [35] show in a clinical setting that miscalibration in either direction carries a cost.

The measurement was single-session and immediate. Each participant rated each response once, shortly after reading it. Trust in a system that a person uses repeatedly develops through accumulated experience of its successes and failures, and a rating given after one exposure is not evidence about that process. Whether the difference observed here would persist, grow or disappear over sustained use is unknown.

Prior experience was not controlled. Every participant was an existing AI-tool user and 80 per cent used such tools daily (Section 4.1.1). Yang et al. [32] report that prior experience and perceived transparency both affect trust independently of actual system reliability, so the ratings reflect a sample already disposed to engage with model output. A sample with no prior exposure might respond differently in direction as well as magnitude.

Taken together, these limitations mean the trust finding should be read as follows: in this sample, a response format that showed its reasoning was rated as more trustworthy than one that did not, by a margin that moved the mean from the disagree region of the scale to the agree region. That is a finding about perception under controlled presentation. It is not evidence that CoT-style responses merit greater trust, nor that participants would rely on them more in practice, and the thesis does not claim either.

5.11 Implications for Human-Centred LLM Design

The findings suggest several implications for the design of human-centred LLM interfaces. Rather than presenting only a final answer, systems could consider providing concise explanatory information when users need additional interpretation or verifi-

cation.

The results indicate that explanatory information can improve users' perceptions of understanding and transparency. However, explanations should not be treated as a substitute for information about reliability or uncertainty. A useful explanation should ideally help users understand the basis of an answer while also communicating uncertainty where appropriate.

This is particularly important for tasks where incorrect information may lead to inappropriate decisions. Users may place greater confidence in a response when it contains a clear and coherent explanation, even when the underlying answer has not been independently verified. Communicating uncertainty and limitations can therefore be important for supporting appropriate reliance on LLM outputs.

For practical LLM interface design, the findings support a human-centred approach in which explanations are clear and useful but are also combined with appropriate signals of uncertainty, limitations, and reliability. Explanations should support users' understanding without creating an unwarranted impression that the model's internal reasoning is fully visible or necessarily correct.

5.12 Discussion Summary

Overall, the study found a consistent difference between the standard and CoT-style response conditions. Participants rated the CoT-style responses higher across all six evaluation dimensions, including perceived understanding, understanding of how the answer was obtained, self-reported trust, confidence, perceived transparency, and explanation clarity and usefulness.

The findings provide support for the value of explanatory information from a user-perception perspective. In particular, participants reported greater understanding of how the answers were obtained and higher perceived transparency when evaluating the CoT-style responses.

However, the results should not be interpreted as evidence that Chain-of-Thought reasoning alone caused these differences. The response conditions also differed in length, detail, information content, and presentation. The study therefore provides evidence about the difference between the two response formats used in the experiment rather than an isolated causal estimate of the effect of CoT reasoning.

The findings for trust should similarly be interpreted as differences in self-reported trust. Higher trust ratings do not establish that the CoT-style responses were objectively more reliable or that the explanations faithfully represented the model's internal reasoning.

Finally, the relationship between perceived explanation quality and trust was positive but not statistically significant. This suggests that explanation clarity and usefulness may be related to trust, but the present study does not provide sufficient evidence to establish a statistically significant relationship.

Taken together, the findings highlight both the potential value and the limitations of reasoning explanations in human–AI interaction. Explanations may help users perceive LLM responses as more understandable and transparent, but their usefulness should be considered together with correctness, faithfulness, uncertainty, and appropriate reliance.

Chapter 6

Conclusions and Future Work

This thesis investigated how Chain-of-Thought (CoT)-based explanations affect users' perceptions of Large Language Model (LLM)-generated responses. The study focused on four aspects of user perception: understanding, trust, perceived transparency, and the relationship between perceived explanation quality and trust. A within-participant evaluation was conducted with 25 participants, where each participant evaluated both CoT-based and standard LLM responses to ten reasoning-related questions.

The results showed a clear difference between the two response conditions in the observed sample. Participants gave higher ratings to CoT-based responses than to standard responses across all six evaluation dimensions considered in the study. These dimensions included ease of understanding, understanding how the answer was obtained, trust, confidence, perceived transparency, and explanation clarity and usefulness. The CoT condition also received a higher overall evaluation score than the standard condition, and the paired statistical analysis indicated a statistically significant difference between the two conditions.

For RQ1, the findings indicate that participants reported greater understanding when evaluating responses that included reasoning explanations. In particular, the CoT condition received higher ratings for both ease of understanding and understanding how the answer was obtained. These findings suggest that, within the conditions of this study, participants perceived the additional reasoning information as useful for understanding the presented answers. However, the study measured self-reported understanding rather than directly observing participants' cognitive processing. Therefore, the results should be interpreted as evidence of perceived understanding rather than evidence of a specific cognitive mechanism.

For RQ2, participants reported higher levels of trust and confidence for the CoT responses than for the standard responses. This indicates that the presence of an explanatory response format was associated with more positive trust-related perceptions in the study sample. However, these results should not be interpreted as evidence that CoT explanations necessarily produce greater or more appropriate trust. Trust can be affected by several factors, including perceived accuracy, prior experience with AI, response presentation, and the amount of information provided to the user.

For RQ3, the CoT responses received higher ratings for perceived transparency than the standard responses. This suggests that participants considered the CoT response format to provide greater insight into how the answer was presented or justified. Nevertheless, the study measured perceived transparency rather than the

actual transparency of the LLM's internal reasoning process. An explanation that appears transparent to a user does not necessarily represent the actual computational process through which the model generated its answer. Therefore, CoT explanations should be understood as a user-facing explanation mechanism rather than direct evidence of faithful model reasoning.

For RQ4, a positive relationship was observed between perceived explanation quality and trust in the CoT condition. However, this relationship did not reach statistical significance. Therefore, the study does not provide sufficient evidence to conclude that higher perceived explanation quality leads to higher user trust. This distinction is important because the higher trust ratings observed for the CoT condition and the relationship between explanation quality and trust represent two different findings. The former describes a difference between the experimental conditions, whereas the latter examines an association between two perception measures. The results of RQ4 should therefore be interpreted cautiously.

Taken together, the results of this thesis support one claim and not another. What the study establishes is that participants rated responses presented with step-by-step reasoning more highly than responses presented without it, on all six evaluation dimensions, consistently across all 25 participants, and by margins that moved every dimension from the disagree region of the scale to the agree region. What the study does not establish is that Chain-of-Thought reasoning alone produced those differences. The two conditions differed in response length, level of detail, structure and formatting as well as in the presence of an explanation, and the design does not separate these. Nor do the results show that CoT-style responses were more reliable, more correct, or more faithful to the model's actual computation; by construction both conditions produced the same final answers. The findings concern how two response formats were perceived, and should be cited as such.

The study also identified several factors that should be considered when interpreting the overall findings. First, the CoT responses contained more explanatory information and were generally longer than the standard responses. Consequently, the present design cannot completely separate the effect of reasoning explanations from the effect of additional information or response length. Participants may have perceived longer and more detailed responses as more informative or effortful, which could have contributed to the observed ratings. Second, the participants had prior experience with AI tools, which may have helped them evaluate the responses but may also have influenced their expectations about explanatory AI outputs.

Another notable characteristic of the results was the consistency of the participant responses. All 25 participants provided a higher overall evaluation for the CoT condition than for the standard condition, and the differences across the six evaluation dimensions were relatively similar. This consistency provides clear descriptive evidence of a systematic difference within the observed sample. At the same time, the uniformity of the responses requires cautious interpretation. The study cannot determine whether this pattern would remain the same with a larger and more diverse population or under different response designs. Factors such as response length, presentation format, participant expectations, and question characteristics may have contributed to the consistency of the observed ratings.

The findings therefore provide support for the value of explanation as a user-facing feature of LLM responses, particularly in relation to perceived understanding and

transparency. At the same time, the results demonstrate that explanations should not automatically be equated with increased reliability or appropriately calibrated trust. This distinction is particularly important because previous research has shown that explanations can influence users' perceptions while not necessarily providing a faithful representation of a model's underlying reasoning [26]. Similarly, research on explanations for end users emphasizes that understandability, trustworthiness, and transparency are related but distinct goals .

Overall, this research contributes a human-centred perspective to the study of Chain-of-Thought prompting. While previous research has primarily investigated CoT in relation to the reasoning performance of large language models , this study examined how users perceive responses generated with and without reasoning explanations. The findings indicate that, in the experimental setting used in this research, CoT-based responses were perceived more positively across the evaluated dimensions. However, the study does not establish that CoT explanations alone caused these differences, nor does it establish that higher explanation quality produces higher trust.

The main implication of this study is therefore that explanation design should be considered an important part of human-centred LLM development. Explanations may help users understand and perceive AI outputs as more transparent, but their design should also consider response length, accuracy, reliability, uncertainty, and the possibility of inappropriate levels of trust. Future LLM systems should consequently aim not simply to provide more explanation, but to provide explanations that are clear, useful, appropriately calibrated, and supported by reliable information.

6.1 Future Work

Several directions can be considered for future research based on the findings and limitations of this study.

6.1.1 Adoption of a Validated Trust Instrument

The most direct methodological improvement available to a replication would be to replace the trust items used here with an established scale. Kohn et al. [30] catalogue the available instruments and the settings each was developed for, which would allow a principled choice rather than the ad hoc items described in Section 3.5. Two benefits would follow. The measurement properties of the trust construct would be known, so internal consistency could be reported rather than left unassessed; and the resulting figures would be comparable with the wider literature on trust in automation instead of interpretable only within this study.

A complementary improvement would be to supplement the self-reported ratings with a behavioural measure of reliance, such as asking participants to decide whether to accept an answer or to verify it, and recording the choice. That would address the first limitation in Section 5.10, namely that the present study measures stated trust rather than acted-upon reliance.

6.1.2 Counterbalanced Presentation and Fuller Demographics

Two changes to the procedure would strengthen a replication at little cost. First, the order of the two condition blocks should be counterbalanced, so that half the participants evaluate the CoT block first. As Section 5.9 explains, the present design presents the blocks in a fixed order and therefore cannot separate the effect of condition from the effect of position. Counterbalancing requires no additional participants and removes the confound entirely.

Second, the background items should record at least gender and age band alongside the existing questions on AI experience. These were not collected here (Section A.1.2), which limits what can be said about how far the sample resembles any wider population, as noted in Section 4.1.1.

6.1.3 Larger and More Diverse Participant Samples

The present study involved 25 participants, which was sufficient to conduct the planned within-participant comparison but limits the generalisability of the findings. Future studies could involve a larger and more diverse participant population, including participants with different levels of AI experience, technical backgrounds, and familiarity with LLM-based systems. A larger sample could also provide greater statistical power for investigating relationships between explanation quality, trust, and other user perceptions.

6.1.4 Comparison of Multiple Large Language Models

This study used the same LLM for generating both response conditions. This design helped control for differences between models and allowed the effect of the prompting condition to be examined more directly. However, future research could compare multiple LLMs to determine whether the observed user perceptions remain consistent across different models. Such a comparison could investigate whether model characteristics influence the clarity, usefulness, or perceived trustworthiness of generated explanations.

6.1.5 Evaluation of Explanation Faithfulness

The present study focused on users' perceptions of explanations rather than whether the explanations faithfully represent the underlying reasoning process of the LLM. Future work could therefore combine user evaluation with technical faithfulness evaluation. This would allow researchers to investigate whether explanations that users perceive as clear and transparent are also faithful to the factors that contributed to the model's output.

6.1.6 Different Explanation Strategies

Future studies could compare CoT explanations with other explanation strategies, such as concise rationales, structured explanations, feature-based explanations, or

explanations specifically adapted to the user's level of expertise. This could help determine whether the benefits observed in this study are specific to CoT prompting or are more generally associated with providing additional explanatory information.

6.1.7 Task and Domain Variation

The present study used reasoning-related questions covering several categories. Future research could investigate whether the effect of explanations changes depending on the task domain. For example, mathematical reasoning, programming, general knowledge, decision-making, and safety-related tasks may require different forms of explanation. Examining multiple domains could provide a better understanding of when explanations are most useful to users.

6.1.8 Long-Term Trust and Reliance

The current study measured participants' reported perceptions immediately after evaluating the responses. Future research could examine trust and reliance over a longer period. Repeated interaction with an LLM could reveal whether the initial increase in perceived trust associated with explanations remains stable or changes as users encounter correct and incorrect AI responses.

6.1.9 Uncertainty and Explanation Design

Future work could investigate whether explanations combined with uncertainty information lead to better calibrated user trust. Rather than simply increasing trust, an effective AI explanation should ideally help users determine when an answer is reliable and when further verification is necessary. Experiments could therefore compare explanations with and without explicit uncertainty information.

6.1.10 Investigating the Relationship Between Explanation Quality and Trust

The present study observed a positive but statistically non-significant relationship between perceived explanation quality and trust. Future studies with larger samples could investigate this relationship in greater depth. Additional variables, such as perceived accuracy, prior AI experience, confidence, and task difficulty, could be included to determine whether they moderate the relationship between explanation quality and trust.

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Appendix A

Supplemental Information

A.1 Survey Instrument

This appendix reproduces the questionnaire as administered through Google Forms.

A.1.1 Introduction and Consent

The following text appeared on the first page of the form, before any item.

> **Evaluation of AI-Generated Responses and Explanations**
>
> This survey is part of a Master's thesis study investigating how different types of AI-generated responses influence users' understanding, trust, and perception of transparency.
>
> You will be asked to review AI-generated responses to several reasoning-related questions and provide your opinions about the responses.
>
> Participation is voluntary. The survey takes approximately 10–15 minutes to complete. No personally identifiable information will be collected, and the responses will be used only for academic research purposes.

A.1.2 Background Items

Three background items were asked, all mandatory. No other demographic information was collected.

1. *How frequently do you use AI tools such as ChatGPT, Gemini, Claude, or similar systems?*
   Options: Daily; Several times a week; Once a week; Occasionally; Rarely.

2. *How would you describe your experience with AI tools?*
   Options: Beginner; Basic; Intermediate; Advanced; Expert.

3. *What is your academic or professional background?*
   Options: Computer Science / IT; Data Science / AI; Engineering; Other technical field; Non-technical field; Other.

A.1.3 Presentation of the Response Material

The following instruction preceded the evaluation items.

> You will be presented with AI-generated responses to several questions. Please read each response carefully and evaluate it based on your own perception. There are no right or wrong answers for the evaluation questions.

The response material was supplied as two PDF documents, one per condition, attached to the questionnaire rather than reproduced inline. The questionnaire then presented two blocks of ten evaluation items each, headed *Response A* and *Response B*. The headings gave no indication of which block corresponded to the standard condition and which to the CoT condition, so participants were blind to the manipulation; the order of the two blocks was the same for every participant, as discussed in Section 5.9.

A.1.4 Evaluation Items

Each of the twenty responses was rated on the same six statements, using a five-point scale labelled 1 – Strongly Disagree, 2 – Disagree, 3 – Neither Agree nor Disagree, 4 – Agree, 5 – Strongly Agree. The item stem read *"Please indicate how much you agree with the following statements about the response you just read."*

1. The response was easy to understand.
2. I clearly understood how the answer was obtained.
3. I trust the answer provided by the AI system.
4. I would feel confident relying on this answer.
5. The response made the reasoning behind the answer transparent.
6. The explanation was clear and useful.

The questionnaire contained no open-ended items. All data analysed in Chapter 4 are therefore from these six statements and the three background items.

A.2 Response Generation Material

This appendix reproduces the material used to generate the two response conditions, so that the stimulus set can be inspected and the generation reproduced.

A.2.1 Prompts

Both prompts opened with the same framing sentence and supplied the question after a list of requirements. The only difference between them lies in those requirements.

Standard condition

```
You are participating in an academic research experiment on AI-generated
responses.
Answer the following question clearly and concisely.
Requirements:
1. Provide the final answer directly.
2. Do not provide a step-by-step reasoning process.
3. Do not provide an extended explanation.
4. Do not introduce information that is not required to answer the
question.
Question: [question text]
```

Chain-of-Thought condition

```
You are participating in an academic research experiment on AI-generated
responses.
Answer the following question and provide a clear, step-by-step explanation
that helps a user understand how the answer was obtained.
Requirements:
1. Explain the reasoning clearly and logically.
2. Present the relevant intermediate steps needed to understand the
answer.
3. Do not introduce irrelevant information.
4. Clearly state the final answer at the end.
Question: [question text]
```

A.2.2 Final Answers in Both Conditions

Table A.1 gives the final answer produced in each condition for each question. The validation described in Section 3.4.4 consisted of confirming that these agreed.

A.2.3 Model and Settings

Three points about this configuration require comment.

First, the responses for the two conditions were generated in two separate sessions, one per condition, with the relevant prompt held fixed across the ten questions within each session. Each question was therefore answered under a single consistent instruction, and no question received an instruction different from the others in its condition.

Second, the four parameters recorded as not available are not omissions in the record but consequences of the access method. The ChatGPT web interface does not expose temperature, top-p, maximum output length or a random seed to the user, and does not report the values it applies. They were therefore neither chosen nor observable, and they cannot be recovered after the fact. Where a published study would normally report these values to permit exact regeneration, this thesis can report only that the interface defaults applied at the time of generation.

Third, the practical consequence is narrower than it may appear. Because large language models are not deterministic at default settings, regenerating the responses from the same prompts would not reproduce identical text even if the parameters

Table A.1: Final answer produced in each condition for the ten experimental questions

ID	Standard condition	CoT condition
Q1	€0.05	€0.05, derived by setting the ball at x and the bat at $x + 1.00$
Q2	No	No; the interns with access need not be researchers
Q3	60 minutes	60 minutes; the tablets are taken at 0, 30 and 60 minutes
Q4	Stress, tiredness, personal concerns, disagreement or having nothing to add; respond by checking in privately, listening without assumptions and offering support	The same conclusion, reached through five steps: consider multiple possible reasons, avoid assuming the cause, respond respectfully during the meeting, check in privately afterwards, and respect the reply
Q5	€135	€135, via €200 → €150 → €135
Q6	Yes	Yes; if training guarantees access, lack of access implies no training
Q7	No	No; timing alone does not establish causation, and training, workload, staffing and technical factors should be considered
Q8	Dependencies, business impact, urgency, effort, resources and risk	The same factors, presented as an ordered set of considerations
Q9	The reason cannot be determined from the information provided	The same conclusion, reached by stating what is given and what is not
Q10	Source credibility, scientific evidence, clinical trials, approval status and professional support	The same factors, presented as six sequential checks

Table A.2: Generation configuration

Property	Value
Model	GPT-5 mini
Access method	ChatGPT web interface
Number of sessions	Two, one per condition
Prompt per session	Fixed for all ten questions within the session
Questions per session	Ten
Responses generated	Twenty in total
Temperature	Not available
Top-p	Not available
Maximum output length	Not available
Random seed	Not available
System message	None set beyond the prompt text
Generation date	Data collection ran from 17 to 22 August 2026, so generation preceded that period.

were known. What the analysis depends on is not the ability to regenerate the material but the material itself, and the twenty responses shown to participants are fixed and published. The two generation sessions are additionally preserved as shared conversation transcripts, so the prompts and the resulting responses can be inspected in the form in which they were produced.

A.2.4 Complete Response Pairs

The twenty responses shown to participants, comprising one standard and one CoT-style response for each of the ten questions, are not reproduced here. Most of the CoT-style responses present five or six reasoning steps, and the full set would extend to several pages of appendix without adding information that a reader would use. Table A.1 gives the final answer each condition produced, which is what the validation described in Section 3.4.4 checked, and Figure 3.2 reproduces one complete pair so that the character of the difference is visible.

The full set is retained by the author as plain text files organised by question and condition, together with the two prompts and the shared transcripts of the two generation sessions, and is available on request as described in the data availability statement. That material is what would be required to inspect exactly what participants read.



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